

kvm-tomcat-lab: complete documentation

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The complete project documentation

Everything in this repository in one document: the plain-English study guide, the full README, every technical write-up, and the raw evidence behind the measured numbers.

Part one: the study guide

1. The story of what we built

Imagine a restaurant

You built a small restaurant. Not a real one - a restaurant made of software, running inside your own laptop. Here is who works there:

In the restaurant	In our project	What it does
The building	Your laptop, with WSL2 and KVM	The place everything happens
Two locked rooms	web01 and app01	Two separate computers inside one
The waiter at the door	Apache on web01	Greets every visitor and passes orders on
The kitchen	Tomcat on app01	Actually makes the food
The recipe	labapp.war	The instructions for the food
The manager with a checklist	Ansible	Walks around making sure everything matches the checklist
A photo you can rewind to	Snapshot	Lets you undo a mistake in seconds
A copy of the recipe at grandma's house	Backup	Saves you if the restaurant burns down
The nurse with a thermometer	Prometheus	Writes down numbers every 15 seconds
The chart on the wall	Grafana	Draws those numbers as pictures
The bouncer with the guest list	Firewall (ufw)	Decides who is allowed through which door

That is the whole project. Everything else is details about how those people do their jobs.

What a big company version looks like

Your restaurant seats 20 people. A company like a bank runs a restaurant with 5,000 tables, in three buildings, in two countries, with a second kitchen that takes over when the first one catches fire.

The workers are the same. There are just more of them, and more rules about what happens when something breaks. **That is why this project is worth building: the ideas are identical, only the size changes.**

What happens when someone visits the page

This is the single most useful thing to understand. Follow one visitor:

1. Someone opens a browser and asks for `http://192.168.122.12/labapp/health`.
2. `.12` is **web01**. The request arrives at Apache, the waiter.
3. Apache looks at its rule sheet (`labapp.conf`) which says: "anything asking for `/`, quietly pass it to `192.168.122.11:8080`". That is a **reverse proxy**: the visitor thinks they spoke to Apache, but Apache just carried the message to the kitchen.
4. `.11` is **app01**, and port `8080` is **Tomcat**, the kitchen.
5. Tomcat looks at the path `/labapp` and thinks "that is the app called `labapp`", which it found in its `webapps` folder.
6. Inside `labapp` the file `WEB-INF/web.xml` says: the address `/health` belongs to the file `health.jsp`.
7. Tomcat runs `health.jsp`, which prints `OK app01` - the name of the machine that cooked it.
8. That text travels back the same way: Tomcat to Apache, Apache to the browser.

The visitor sees `OK app01`. Notice the name says **app01**, even though they asked **web01**. That is the proof the request really travelled between the two machines instead of being answered at the front door.

Analogy. You order soup. The waiter does not cook it, but they do hand it to you. If the bowl has the kitchen's stamp on it, you know the kitchen really made it.

Why bother with two rooms?

A fair question. One computer could be both the waiter and the kitchen.

We split them because that is what real systems do, and each half gives you something:

- **One front door.** Visitors only ever see `web01`. The kitchen is hidden, so nobody can walk in and start poking the ovens. We proved this: after hardening, knocking directly on the kitchen door (`app01:8080`) from outside gets **no answer at all**.
- **One place to add security.** When you eventually add HTTPS, you do it once, at the front door. The kitchen never needs to know about certificates.
- **One place to write the guest book.** Every access log is in one file on `web01`, instead of scattered across machines.
- **Room to grow.** If the kitchen gets busy you can add a second kitchen later and have the waiter share orders between them. If the waiter gets busy, add a second waiter.
- **Smaller blast radius.** If the front door breaks, you can still reach the kitchen to fix it - and the other way round.

Analogy. A restaurant where customers walk straight into the kitchen is a food truck. Fine for one customer, chaos for a hundred.

The order we built it in, and why that order

The build happened in nine steps, called **modules**. The order is not random.

Step	What we did	Why it had to come in this position
1	Set up the machine that makes machines (KVM)	Nothing else exists until something can create a computer
2	Made two computers with fixed addresses	Ansible needs to know where they are, by address
3	Wrote the tiny app	The kitchen needs a recipe before it can cook
4	Told Ansible to build everything, and ran it twice	Twice is the proof it is safe to repeat
5	Took a photo (snapshot), broke things, rewound	Learn the undo button before you need it
6	Made a backup and restored it from scratch	Learn the ambulance before the accident
7	Put instruments on and ran a load test	You cannot tune an engine you cannot measure
8	Locked the doors	Do this last: a firewall mistake can lock you out
9	Wrote it all down	If it is not written down, it did not happen

Rule of thumb worth keeping: learn the undo and the safety net before you start breaking things on purpose. Module 5 and 6 before 8 is not an accident.

The numbers we ended up with

Short version, because Chapter 5 explains each one:

- Running the setup tool twice in a row: the second time it changed **nothing** (`changed=0`). That is the proof it is safe to run again and again.
- The site answers through the front door, and the kitchen door is **closed** to outsiders.
- The backup was deleted, then restored from the copy kept elsewhere, and every fingerprint matched: **RESTORE VERIFIED: 5 files match**.
- Deliberately corrupting a backup made the tool **refuse** it and exit with an error. A checker that never complains is not a checker.
- Adding processors: **1 → 2 → 4 processors** gave **17.9 → 29.2 → 44.1 requests per second**. Four times the cooks, 2.46 times the soup - and Chapter 5 explains why that is a good result, not a disappointing one.
- The restaurant was completely destroyed (the whole computer deleted) and rebuilt, serving again in **5 minutes 48 seconds**.

How to read the rest of this guide

Every subject below is explained the same four ways, so you can skip around:

- **What it is** - the plain idea, usually with an analogy.
- **Why we needed it** - the actual problem it solved in this project.
- **How we used it** - the concrete thing that happened here.
- **Where you will meet it again** - the real jobs and systems that use it.

Chapter 7 is a glossary and a command cheat sheet. If a word ever loses you, jump there.

2. Machines inside machines, and the wires between them

This chapter is about the ground floor: how you made two computers without buying two computers, and how they talk to each other.

A computer inside a computer (virtualization)

What it is. A program that pretends to be a whole computer. You give it some of your memory, some of your disk and some of your processors, and it behaves like a separate machine with its own operating system. You can start it, stop it, break it, and throw it away without touching the real computer.

Analogy. A folded paper house. It is a real house - you can live in it, put furniture in it - but it is standing inside a bigger room. Knock it down and the big room is fine.

Why we needed it. The project is about running servers. Real servers get installed, changed, backed up and destroyed. Doing that on your own laptop directly would be dangerous and messy. Virtual machines make it safe to experiment.

How we used it. We made two virtual machines: `web01` (the waiter) and `app01` (the kitchen).

Where you will meet it again. Almost everywhere. Any “cloud server” you rent is a virtual machine. Any company server room runs hundreds of them on a handful of big physical machines, because most servers sit idle and it is wasteful to give each one its own hardware.

The word for the boss program. The thing that creates and manages virtual machines is called a **hypervisor**. Three famous ones: **KVM** (what we used, free, part of Linux), **VMware vSphere** (what big companies often pay for), and **Hyper-V** (Microsoft’s). They all do the same job with different buttons.

KVM, qemu, and libvirt - three names for one job

Beginners get confused here because three programs are involved. Here is the division of labour:

Name	Plain meaning	In our project
KVM	The part built into Linux that lets a virtual machine use the real processor at full speed	It is a feature of the Linux kernel. We checked it was switched on.
qemu	The program that pretends to be the hardware: the disks, the network card, the screen	One <code>qemu</code> process runs for each virtual machine
libvirt	The manager. It keeps a list of machines, remembers their settings, and gives you one tidy set of commands	The <code>virsh</code> command you typed talks to libvirt
virsh	The command you type to talk to libvirt	<code>virsh list --all</code> shows your machines
virt-install	A helper that builds a new virtual machine in one command	<code>scripts/create-vms.sh</code> uses it

Analogy. KVM is the engine, qemu is the car body and wheels, libvirt is the garage manager who knows where all the cars are parked and what keys they need, and `virsh` is the clipboard you talk to the manager with.

Why not just use qemu directly? You can, but you would have to write a 40-line command every time and remember dozens of options. libvirt remembers for you, and gives you useful extras like snapshots and live resizing.

Where you will meet it again. `virsh` skills transfer directly. If you learn `virsh list`, `virsh start`, `virsh dominfo` and `virsh snapshot-list`, you will recognise the same shapes in VMware (`vim-cmd`, PowerCLI) and Hyper-V (`Get-VM`).

Why we ran Linux inside Windows (WSL2)

What it is. WSL2 is Microsoft's way of running a real Linux system inside Windows, sharing the same computer. It is not a pretend Linux; it is actual Linux in a lightweight virtual machine that Windows starts for you.

Analogy. A guest flat built into your house. Separate front door, separate kitchen, but the same building and the same plumbing.

Why we needed it. Your laptop runs Windows. KVM exists only in Linux. So we needed a Linux machine, and instead of installing one or buying a second laptop, WSL2 gave us one in about two minutes.

The important trick: nested virtualization. This means running a virtual machine inside a virtual machine - our KVM machines live inside the WSL2 machine. Windows has to agree to pass the processor's virtualization feature down the chain. It did, and we proved it:

```
sudo kvm-ok
# INFO: /dev/kvm exists
# KVM acceleration can be used
```

That `/dev/kvm` file is the doorway to the processor's virtualization feature. If it is missing, virtual machines still run but in **software emulation** - the computer pretends to be a different processor in slow motion. Everything works, just 10 to 50 times slower.

Honest cost. We are running inside a wrapper, so speeds are lower and the guest machines are capped by whatever WSL is given. The project's absolute speed numbers describe a laptop, not a data centre. Chapter 5 says more about why that still makes the experiment useful.

Where you will meet it again. WSL2 is extremely common now for developers on Windows. On servers you would use real Linux, but every command you learned here is the same command.

A ready-made computer: the cloud image

What it is. A file containing a complete, already-installed Linux system. Not an installer - a finished disk. You copy it and boot it, and 20 seconds later you have a working computer with no setup questions.

Analogy. A frozen meal versus cooking from scratch. Making Linux from an installer takes 20 minutes of clicking. The frozen version is ready in seconds, and you add your own salt afterwards.

Why we needed it. We needed to build the same machine twice - once normally, and once again during the disaster recovery test. Doing that with an installer would be slow and error-prone. With a cloud image it is one command.

How we used it. We downloaded Ubuntu's cloud image (597 MB) and **checked its fingerprint** against the publisher's published list before using it. That fingerprint check matters more than it looks: a corrupted or tampered base image is a problem you would not notice for months.

Where you will meet it again. Every cloud provider starts from images like this. "Amazon Machine Image" and "Azure Marketplace image" are the same idea.

cloud-init: the note you leave for a new computer

What it is. A way to hand a brand-new machine a small note on first boot saying: your name is this, make this user, here are the keys, turn off password logins.

Analogy. A note taped to the fridge in a new flat: "Your name is app01. The spare key belongs to Ziaur. Do not let anyone in without a key."

Why we needed it. A fresh cloud image has no user, no password and no way in. Without cloud-init you would have to open a console and type everything by hand for each machine.

How we used it. Our note created a user called `ops`, gave it permission to run administrator commands without a password (so Ansible can do its job), and installed our SSH key.

Where you will meet it again. Every cloud. It is the standard way to give a new server its identity. On the job you will write these notes, called `user-data` files, all the time.

Thick disks that are secretly thin (qcow2 and backing files)

What it is. A disk file format that only takes up space for the data you actually write. A “10 GB disk” may only use 700 MB on your real disk. It can also be stacked: a small disk on top of a big read-only one.

Analogy. Tracing paper. Instead of copying a whole drawing, you lay a sheet of tracing paper over it and only draw your changes. The original never gets touched, and your sheet is nearly empty.

Why we needed it. Two virtual machines each wanting a 10 GB disk would mean 20 GB of real space, most of it empty. With stacking, the 597 MB base is stored **once**, and each machine has a thin layer of its own.

How we used it. `create-vms.sh` does exactly this:

```
qemu-img create -f qcow2 -F qcow2 -b noble-server-cloudimg-amd64.img app01.qcow2 10G  
  
#                ^^ the base image                ^^ the thin sheet on top
```

Where you will meet it again. VMware calls it a **linked clone**, Hyper-V calls it a **differencing disk**, and Docker calls it a **layer**. Same idea, three brand names. This is one of the most transferable concepts in the project.

The ceiling you set when you build the machine

What it is. When you create a virtual machine you say two numbers: how much it has now, and the maximum it is ever allowed to have.

Why we needed it. Because “add a processor to a running machine” only works up to a limit fixed at creation. We wanted the machine to start small (so the load test would find a real bottleneck) and still be able to grow later.

How we used it. `app01` was created with **1 processor now, maximum 4**, and **2 GB of memory now, maximum 4 GB**. When we later tried to be greedy, the computer said no, politely and clearly:

```
$ virsh setvcpus app01 5 --live
error: requested vcpus is greater than max allowable vcpus for the live domain: 5 > 4

$ virsh setmem app01 8G --live
error: cannot set memory higher than max memory
```

Where you will meet it again. VMware and Hyper-V have the same concept - memory reservations and limits set when the machine is built. Interviewers like this question because it shows whether you have actually done it or only read about it.

The wires: a private network just for the machines

What it is. libvirt creates a small private network for its machines: `192.168.122.0/24`. Several things live in that one line.

Analogy. A private road behind the houses. The delivery van (the internet) can reach the road through the gate, and the houses can talk to each other easily.

Piece	Plain meaning	In our project
NAT network	Machines can reach the internet, the internet cannot reach in	Lets the machines install software
Bridge (<code>virbr0</code>)	The virtual switchboard the machines plug into	The thing that broke every time WSL restarted
DHCP	The automatic address giver	Hands out addresses when machines boot
DHCP reservation	"This machine always gets this address"	How <code>app01</code> is always <code>.11</code>
Gateway	The way out of the private road	<code>192.168.122.1</code> , the host itself

Why fixed addresses matter. Ansible's address book contains two addresses. If a machine woke up with a different one, every instruction would be sent to an empty house. The reservation is made against the machine's **MAC address** - the hardware's permanent serial number for its network card - so it does not matter what order machines start in.

Where you will meet it again. Every network you ever touch. Subnets, NAT, DHCP, gateways, MAC addresses - these concepts are in routers, cloud networks and office networks alike.

Getting in without a password (SSH keys)

What it is. A way to prove who you are using two matched files instead of a typed password: a **private key** you keep secret, and a **public key** you give away. The machine keeps a copy of your public key; if you can prove you hold the matching private key, you are let in.

Analogy. A padlock you hand out and the only key that opens it. You can give the padlock to a thousand people and they still cannot open it.

Why we needed it. Ansible logs in dozens of times per run. Typing a password each time is impossible, and passwords can be guessed. Keys cannot reasonably be guessed, and they let automation work.

How we used it. One key pair for the lab. The public half was installed by cloud-init; password logins were switched off entirely. We then proved it:

```
$ ssh -o PubkeyAuthentication=no ops@192.168.122.11
ops@192.168.122.11: Permission denied (publickey).
```

Where you will meet it again. Everywhere in servers and code hosting. The same idea is behind GitHub's push access, and behind every automated deployment you ever write.

The rule you must never break: the private key file never goes into the project folder and never into git. Our project automatically ignores files named like keys, and we checked that no key was ever committed.

3. The kitchen, the waiter, and the bouncer

Chapter 2 built two computers. This chapter puts food in one and a doorman on the other.

Java: the language the kitchen speaks

What it is. A programming language with one special trick: the same program can run on Windows, Linux or a Mac without being rewritten.

Analogy. Esperanto for computers. You write the recipe once, and every kitchen in the world can read it.

Why it matters. Java code does not run directly on the processor. It runs inside a program called the **JVM** (Java Virtual Machine), which translates as it goes. That middle layer is why Java is slower than some languages and why it works everywhere.

Where you will meet it again. Enormous amounts of business software are Java: banking, insurance, government systems. If a job mentions Tomcat, WebLogic, WebSphere, Spring or JBoss, the language underneath is Java.

Tomcat: the kitchen itself

What it is. A **web application server** for Java. It knows how to receive web requests, find the right piece of Java code, run it, and send the answer back.

Analogy. A kitchen with a rule: “Put your recipe box on this shelf and I will cook from it.” You do not hire cooks; you hand Tomcat a box and Tomcat cooks.

Why we needed it. Something has to actually run our code in response to a browser asking. Tomcat does that, plus a lot of unglamorous but essential work: managing many visitors at once, restarting a broken app, writing logs.

How we used it. We installed the `tomcat10` package, and Tomcat watches one folder: `/var/lib/tomcat10/webapps/`. Anything placed there gets picked up automatically.

Where you will meet it again. Tomcat is the most common Java application server in the world, and it is free. Its big commercial cousins are WebLogic (Oracle) and WebSphere (IBM) - same job, more buttons, much bigger price. If you can explain Tomcat, you can explain those in an interview.

JSP: a web page with code inside it

What it is. A file that looks like a page of text but can contain small bits of Java. Tomcat turns it into real program code the first time somebody asks for it.

Analogy. A recipe card with a note in the margin: “write today’s date here”. The card is mostly fixed text; the margin changes each time.

How we used it. Two tiny files:

```
health.jsp    -> prints: OK app01
work.jsp      -> spends about 87 milliseconds doing heavy arithmetic, then says
"done"
```

Why the slow one? Because of Chapter 5. To learn where a system gets stuck you need something that genuinely works the processor hard. A page that just prints text takes 1 millisecond and teaches you nothing. `work.jsp` was written to be deliberately expensive, and we tuned it until one request cost about 87 milliseconds - in the range a real, slightly heavy web page might cost.

A detail worth knowing. The very first request to a JSP is slow, because Tomcat has to compile it. That is why our Ansible task waits and retries instead of expecting success immediately. If you have ever wondered why a site is slow the first time and fast afterwards, this is often the reason.

Where you will meet it again. JSP is old-fashioned now - modern Java web apps use Spring Boot with templates - but the idea (a page with placeholders filled in by code) is everywhere. It is the same idea as PHP, or as templates in Python and JavaScript.

WAR: the recipe box

What it is. A single file containing a whole web application: pages, code, settings, all zipped up. The name means “Web Application aRchive”.

Analogy. A lunchbox. Everything the kitchen needs, in one box, with the label on the outside.

Why we needed it. It gives us one object to build, one object to copy, and one object to back up. That simplicity is the reason the whole deployment is a single step in Chapter 4.

How we used it. `app/build.sh` zips three files into `labapp.war`, and the file name decides the web address: `labapp.war` becomes the address `/labapp`. Tomcat unpacks it into a folder of the same name.

The little bug we found. Building the same source twice produced two different files. That sounds harmless but it broke our proof that the setup tool changes nothing on a second run - because the tool kept seeing “a different file!” and copying it again. The cause was extra hidden metadata that `zip` writes into archives. One extra flag (`zip -X`) fixed it. Chapter 6 tells the full story, because the lesson is bigger than the bug.

Where you will meet it again. WAR files are still how Java apps are shipped to Tomcat, WebLogic and WebSphere. Java’s newer sibling is the JAR (a plainer box), and other languages have their own versions: a `.whl` for Python, an `.npm` package for JavaScript. Same idea: one file, everything inside.

Apache: the waiter at the door

What it is. A **web server**: the program that receives web requests first. It is very good at the simple, high-volume work of greeting, checking, logging and handing requests onward.

Analogy. A restaurant's front-of-house staff. They do not cook, but they greet you, take your order, keep the guest book, and know that the kitchen is through the swing doors.

Why we needed it - and not just Tomcat? A fair question, and interviewers ask it. Four real reasons:

1. **A single entrance.** Visitors meet one machine, not several. From outside, the system looks simple.
2. **Security in one place.** When you eventually add HTTPS, you configure it once here. The kitchen never handles certificates.
3. **One guest book.** All access records in one file, which makes investigating problems much easier.
4. **Growing sideways.** If you add a second kitchen later, the waiter can share orders between them. Tomcat alone cannot do that.

reverse proxy: carrying the order to the kitchen

What it is. A rule that says "requests arriving here should quietly be handed to that other machine over there, and the answers brought back". The visitor never knows it happened.

Analogy. A phone answering service for a small business. You dial one number; a friendly voice answers and passes your message to whoever actually does the work.

How we used it. Two lines inside a virtual host file - the shortest and most important lines in the whole project:

```
ProxyPass          / http://192.168.122.11:8080/  
ProxyPassReverse   / http://192.168.122.11:8080/
```

Read it as: "anything asking for / should go to 192.168.122.11 on port 8080". The second line fixes up the answers on the way back so links work properly.

A virtual host is just a section of Apache's settings for one website. We disabled Apache's default one so only ours is active.

Where you will meet it again. Every serious website does this. The same job is done by nginx, HAProxy, and cloud load balancers. The term for the general idea is a **load balancer** when there are several kitchens, and a **reverse proxy** when there is one.

The bouncer: a firewall decides who may knock

What it is. A list of rules about which doors are open, to whom. Anything not allowed is ignored completely - the packets are dropped, so a visitor gets silence rather than a refusal.

Analogy. A bouncer with a guest list. Not on the list? You do not get told why; you simply do not get in. The door behind the building that everyone forgot about gets locked too - that is the part that matters.

Why we needed it. Before this step, the whole world (well, our little network) could knock directly on the kitchen door and skip the waiter. That made the waiter **decorative**. The project claimed a two-tier design, but nothing enforced it.

How we used it. We installed `ufw` (“uncomplicated firewall”) and wrote rules that match the design exactly:

On <code>app01</code> (the kitchen)	Who may knock
port 22	anyone (so we can administer it)
port 9100	only the monitoring machine
port 8080	only <code>web01</code> - the waiter

On <code>web01</code> (the waiter)	Who may knock
port 22	anyone
port 9100	only the monitoring machine
port 80	anyone - the only public entrance

Proof it worked. Before: knocking straight on the kitchen door returned “Hello, here is your soup” (HTTP 200). After: nothing at all.

```
through Apache:      HTTP 200  body=OK app01
app01:8080 from host: HTTP 000 (curl exit 28)
```

Exit code 28 means “timed out” - the packets were dropped, not refused. And we checked the other side too: the kitchen logged 10 blocked-connection entries. **A door that silently does not open is impossible to diagnose; a door that logs why is a door you can fix.** That detail is worth more in an interview than the rules themselves.

The dangerous part. A firewall mistake can lock you out of the machine you are working on. Our rules are written so all the “may come in” lines happen before the “everyone else is blocked” line. Reverse that order and the first thing the firewall blocks is the connection running the job that is configuring it. We also made sure a snapshot existed, and knew the console command to get back in, before running it.

Where you will meet it again. Constantly. Cloud providers call the same thing a **security group** or a **network ACL**, and they expect you to know which port needs

opening for which service. “Why can’t the servers talk to each other” is one of the most common problems in operations, and this is usually the answer.

4. Doing it automatically, and protecting it

Chapters 2 and 3 built the restaurant by hand. This chapter is about the manager with the checklist, the alarm clock, the photograph you can rewind to, and the copy of the recipe kept at grandma's house.

Why not just type commands like normal?

You could. Here is what happens when you do, three months later, on a second machine:

- You forget one of the 20 steps.
- You type it slightly differently, and something behaves slightly differently.
- You cannot tell whether the machine is correct or whether you just got lucky.
- You cannot show anyone what you did.

So instead of typing commands, we describe the **result we want** in a file, and let a tool make it true. The tool is Ansible.

Ansible: the manager with a checklist

What it is. A program that logs into machines and makes them match a written list. You write what you want - "Tomcat installed, this file here, this service running" - and Ansible works out the commands.

Analogy. A housekeeper with a checklist. You do not say "walk to the cupboard, pick up the cloth, move your arm in circles". You say "the kitchen should be clean". How to achieve it is their problem.

What is in the checklist, and the vocabulary:

Piece	Plain meaning	Our example
Playbook	The whole checklist	<code>site.yml</code>
Inventory	The address book of machines	<code>inventory.ini</code>
Role	A group of jobs for one purpose	<code>tomcat</code> , <code>apache</code> , <code>backup</code>
Task	One line on the checklist	"Install Tomcat 10"
Handler	A job that only runs if something actually changed	"restart Apache, but only if you edited its settings"
become	Run as the administrator	Needed to install software

Why we needed it. Two machines, dozens of settings, and a house rule that we must be able to rebuild everything after a disaster. Typing is not an option.

Why "the file decides the address" is a good thing. Read our inventory:

```
[app]
app01 ansible_host=192.168.122.11
```

Every instruction for the kitchen is written against the name `app01`. That is why the fixed address from Chapter 2 matters: names point at addresses, and if the address drifted, the instructions would go nowhere.

Where you will meet it again. Ansible is one of the most requested skills in operations job adverts. Its rivals are Puppet, Chef and Salt - same idea, and the concepts move across easily. Small setups often use plain scripts instead, but the moment there are more than a couple of machines, people reach for a tool like this.

Idempotent: the most important idea in the project

What it is. A long word for a simple property: **doing it twice is the same as doing it once.**

Analogy. A light switch, not a doorbell. Press a light switch twice and the light is on. Press a doorbell twice and you have rung it twice. Configuration should behave like the light switch.

Why it matters so much.

- You can run the checklist again without fear. It is always safe.
- You can run it after an interruption without unpicking what half-happened.
- You can use it to check a machine: run it, and if nothing changes, the machine is correct. That is a health check for free.
- It is the difference between “a script I ran once” and “a system I can manage”.

How we proved it. We ran the playbook twice and looked at the summary:

```
PLAY RECAP
app01  : ok=18   changed=0    failed=0      <- ran 18 jobs, changed nothing
web01  : ok=17   changed=0    failed=0
```

Read that carefully, because there is a subtlety worth understanding. `ok=18` with `changed=0` does **not** mean the jobs were skipped. It means all 18 jobs ran, looked at the machine, and reported “this is already how you asked it to be”. `failed=0` means none of them broke.

If a job is skipped you see `skipped`, which is a different thing entirely. If a job does something you see `changed=1`. **`changed=0` is the proof.**

A trap we fell into. Our proof was quietly false for a while, because one job kept reporting `changed`: we were rebuilding the app file, and the rebuild produced a slightly different file every time. So Ansible honestly said “this file is different, I copied it again”. The checklist was fine; the thing we handed it was not stable. Chapter 6 has the full story.

Where you will meet it again. In every interview about automation, and in every real system. The word, the idea and the proof are all expected knowledge.

Git and GitHub: a notebook that remembers everything

What it is. Git keeps the entire history of a folder - every change, who made it, and why. GitHub is a website that holds a copy of that history so others can see it, and so you can work on it from anywhere.

Analogy. A lab notebook where you cannot erase. You can write a new page saying “actually, yesterday’s page was wrong”, but yesterday’s page stays there for anyone to read. That is a feature, not a bug.

Why we needed it. Three copies of safety: the project can be rebuilt from it (which is exactly what the disaster recovery test proved), the history explains why things are the way they are, and it is the thing a future employer looks at.

The workflow we used, and why it looks fussy. Every piece of work went on its own **branch** and came back through a **pull request**:

1. Make a branch - a private copy of the work, like a rough draft.
2. Do the work, in several small saved steps (**commits**).
3. Ask for it to be added to the main version (**pull request**, or PR).
4. Look at the change one last time, then merge it in.

Analogy. Nobody redraws the class poster by rubbing bits out while everyone watches. You draw your idea on a separate sheet, show it, and only then stick it on the poster.

Even working alone, this helps: the PR is a pause where you look at your own work as a stranger would. It is also exactly how every software team works, so practising the shape now means it is familiar later.

A small habit worth copying. Every commit message says what changed and why, not just “update”. Three months from now, “why” is the only part you will need.

systemd timers: the alarm clock

What it is. Linux’s built-in scheduler. You describe a job and when it should happen, and Linux runs it for you.

Analogy. An alarm clock next to a note that says “when the alarm goes off, make a backup”.

Why we needed it. A backup you have to remember is not a backup. Ours runs itself every night at 2am.

Two units, not one. The note (`.service`) says what to do; the alarm (`.timer`) says when. That separation is a Linux convention.

A detail worth understanding. We set `Persistent=true`, which means: if the machine was switched off when the alarm should have rung, run the job as soon as it wakes up. Our

lab machines are frequently off, so without this a night's backup would silently not happen.
Silently not happening is the worst behaviour a backup can have.

Where you will meet it again. Timers and their older cousin **cron** are on every Linux server. Anything that must happen regularly - backups, cleanups, reports, certificate renewals - runs this way.

tar and gzip: the box

What it is. `tar` packs many files into one file. `gzip` squeezes that file to make it smaller. Together they make `.tar.gz`, one tidy box.

Analogy. Putting the whole kitchen's recipe cards into one box, then vacuum-sealing the box.

Why we needed it. One object is far easier to copy, verify, move and store than a folder of loose files.

Where you will meet it again. Constantly. Nearly every backup, download and software package you handle is a `.tar.gz` underneath. (`.zip` is the Windows cousin and does the same job.)

SHA-256: the fingerprint

What it is. A recipe that turns any file into a short string of letters and numbers - its fingerprint. Change one letter in the file and the fingerprint changes completely. You cannot work backwards from the fingerprint to the file.

Analogy. A fingerprint. It says that this is the same person, without telling you anything else about them. Cut your hair and the fingerprint still matches; change your identity and it will not.

Why we needed it. Fingerprints are how we answer the question "is this exactly the same as that?", which is the heart of every check in this project. Two places in particular:

- **Is the backup really the same as what we backed up?** Compare fingerprints.
- **Did this file survive the trip to another machine?** Compare fingerprints.

Where you will meet it again. Everywhere in modern computing. Downloads, code hosting, package managers, blockchains, git itself. The name is worth understanding: SHA-256 is one specific fingerprint recipe, and it is the common one.

A manifest: the list of fingerprints

What it is. A file listing the fingerprint of every file inside a backup.

Analogy. An inventory list taped to the box, saying "inside: 5 items, and here is each one's fingerprint".

Why we needed it. Without it, a backup is a box you have never opened. With it, you can prove the contents later, on a different machine, without having the original to compare against.

How we used it. Our backup tool writes this list as it packs the box. Later questions like “does this restore contain the right things?” become a simple comparison of two lists.

Where you will meet it again. Software vendors publish checksum lists so you can check a download. Package managers keep manifests of installed files. It is the same trick.

The backup that checks itself

This is the part of the project worth the most in an interview, and it is only about 30 lines of shell script. Here is the pattern:

```
pack the folder into a box
-> open the box into a temporary scratch folder
-> fingerprint everything found inside
-> compare against the list of fingerprints taken from the original
-> different? shout "RESTORE MISMATCH" and exit with an error
```

Why not just fingerprint the box itself? Because that only proves the box did not rot while sitting on the shelf. It proves nothing about whether the box contains your application. Opening it and comparing the contents answers the real question: if I unpacked this on a brand-new machine, would I get my app back?

Why the failure drill matters. A checker that only ever prints “fine” is useless. So we deliberately corrupted a backup and demanded the tool refuse it. It did:

```
RESTORE MISMATCH: /var/backups/tomcat/webapps-...tar.gz
exit code: 1
```

Then we put it back and demanded it pass again - so we know the tool can tell the difference, rather than always complaining.

Where you will meet it again. “We have backups” is a claim. “We have restored a backup this quarter” is evidence. Employers have been burned by the first kind of claim, so the second kind gets you hired.

rsync: the careful mover

What it is. A tool for copying files between machines that only transfers what has changed, and can pick up where it left off.

Analogy. A removal van that checks which boxes are already at the new house and only carries the rest.

Why we needed it. A backup that lives only on the machine it protects is not a backup - it dies with the machine. We copy the boxes to the host machine and **check the fingerprints again after the journey**, because a copy that silently went wrong is worse than no copy, since you think you are safe.

The honest limitation. Our copy sits on the same physical laptop. It survives the virtual machine dying - which the disaster recovery test proved - but not the laptop dying. Real systems send copies to a different building or to cloud storage. Saying that out loud is part of being trustworthy about a system.

Where you will meet it again. rsync is one of the most-used tools in Linux operations. Its cousins are `scp` (plain copy over SSH) and cloud storage tools.

Snapshot versus backup: the photograph and the copy

This is the question interviewers ask most often, so learn the difference properly.

	Snapshot	Backup
Analogy	A photograph of the room	A copy of the recipe kept at grandma's house
Where it lives	On the same disk as the machine	Somewhere else
Good for	Undoing a mistake in seconds	Surviving a disaster
Survives losing the disk?	No - it dies with it	Yes, that is the whole point
Contains memory?	Optionally yes, as ours did	No, it is a file archive
Is it checked?	Not applicable	Restored and fingerprinted

How we proved the difference instead of just claiming it. In the disaster recovery test we deleted the machine with its disk. The snapshot went too. Nothing about the snapshot could have saved us; only the backup could.

The snapshot drill we ran. Take a photograph, then deliberately wreck the kitchen, then rewind:

- Deleting the app: the site went to **404** ("the kitchen is fine, but the recipe is missing").
- Rewinding: **5.1 seconds**, and the app came back with the identical fingerprint.
- The clock test: the machine's uptime read "up 1 minute" before the rewind and "up 1 minute" after it, proving the rewind did **not** reboot the machine - it restored a running moment. That is why it takes five seconds instead of the three minutes a boot costs.

Reading error numbers like a detective. 404 and 503 look similar and mean different things:

- **404** = the kitchen is alive and well, but the recipe is missing.
- **503** = the kitchen itself is not answering.

Knowing which one you are looking at saves an hour of hunting in the wrong place. This exact distinction cost us time during the build, in Chapter 6.

Growing a running machine (live resize)

What it is. Adding memory or processors to a machine that is switched **on**. No reboot, no downtime.

Analogy. Laying another table for a dinner party while the guests are eating, rather than asking everyone to leave and come back.

Why we needed it. Because it is how real capacity planning works: you measure, then you add, then you measure again. And because the maximum you are allowed to grow to must be set when the machine is created (Chapter 2).

Memory worked instantly. The machine saw the new memory immediately, with no reboot at all: 1867 MB became 2891 MB inside the guest.

The processor arrived but stayed asleep. This is the best small discovery in the project. libvirt said “you now have 2 processors”. The machine still said “I have 1”. The machine’s own log said:

```
ACPI: CPU1 has been hot-added
```

So the processor was delivered and plugged in - and then left switched off. The number you check (`nproc`) counts processors that are awake. One extra command to switch it on, and `nproc` said 2.

Why this is worth telling an interviewer. It shows the difference between “the tool said it worked” and “I verified the thing actually works”. Many people stop at the first one.

Where you will meet it again. VMware calls it hot-add, Hyper-V has dynamic memory. The same surprises happen, because they come from the same place: the guest operating system has to notice and accept the change.

5. Watching, measuring, and finding the slow part

Chapter 4 protected the restaurant. This chapter puts instruments on it and works out where the queue forms.

Why measure at all?

What it is. Writing down numbers about a system over time, so you can answer questions instead of arguing about opinions.

Analogy. A thermometer and a chart on the kitchen wall. Without them, “the kitchen feels slow today” is the best anyone can say. With them, you can see that the kitchen hits its limit every day at noon, and does nothing about it for the other 23 hours.

Why we needed it. Three jobs:

1. **To see problems before people complain.** “Is the machine about to fill its disk?” is answerable in advance.
2. **To find the slow part.** Guessing where a system is stuck is usually wrong.
3. **To prove a change helped.** We added processors and could show exactly what it bought us, in numbers.

Where you will meet it again. Every operations job. The general term is **observability**, and it splits into three related ideas worth knowing by name: **metrics** (numbers over time), **logs** (things that happened), and **traces** (the path one request took through a system).

Prometheus: the nurse with the notebook

What it is. A program that asks every machine “how are you?” every few seconds, writes down the answers with a timestamp, and lets you ask questions about the history.

Analogy. A nurse who walks the ward every 15 seconds and writes down everyone’s temperature. Nobody has to be watched; the notebook fills itself.

Why we needed it. Somebody has to collect the numbers, and importantly, keep them. A number you looked at once is forgotten. A number written down every 15 seconds lets you see that the processor has been busy since 9am.

Two words worth knowing. Each machine “answers questions” at a web address - ours at port 9100. Prometheus **scrapes** that address, which simply means it fetches the numbers on a timer.

Where you will meet it again. Prometheus is the standard free monitoring system. Its hosted cousins are Datadog, New Relic and Grafana Cloud. The words “scrape”, “metric” and “alert rule” transfer directly.

node_exporter: the little meter on each machine

What it is. A small program installed on a machine that reports basic facts about it: how busy the processor is, how much memory is free, how full the disk is, how much network traffic there has been.

Analogy. A meter reader for a house. It does not fix anything; it just knows how much electricity is being used and tells anyone who asks.

Why we needed it. Prometheus cannot see inside a machine by itself. It needs something inside to report. That is all `node_exporter` is.

A useful detail. It is not part of the operating system; it is just a program. If it stops, monitoring of that machine stops - so one of our alert rules is literally “this machine has stopped answering”.

Where you will meet it again. Almost every Linux server being monitored has one of these installed. The pattern (a small agent reporting local facts) repeats everywhere: container agents, database exporters, application exporters.

A metric: a number with a timestamp

What it is. The basic unit of monitoring: a number, a name, and the time it was taken.

Analogy. A row in the nurse’s notebook: “3:15pm - Mr Patel - 37.1”.

Why it matters that numbers come with timestamps. Because a single number is almost meaningless and a series of them is a story. “Processor at 100%” is scary. “Processor at 100% for two minutes and then back to 5%” is a build job. The history is the information.

PromQL is the little language for asking questions of these numbers. Our whole alerting setup uses expressions a person can read aloud, for example:

```
100 - (average idle processor percentage) > 80
```

That reads as: “if the machine has been more than 80% busy for two minutes, tell me”. The “for two minutes” part matters - a brief spike is normal, so rules wait before shouting. Learning to write that short delay is the difference between alerting that people trust and alerting that people ignore.

Where you will meet it again. Every monitoring system has a query language. Learning one teaches you the shape of them all.

Alert rules: the rules that decide when to say something

What it is. Small statements that turn numbers into meanings: “if this, for this long, then that is a problem”.

Analogy. A smoke alarm. Not a fire - a rule about fires, so nobody has to sit and watch.

Why we needed them. Nobody watches a chart for a living. Rules watch the chart and only bother a human when something is genuinely wrong.

How we used them. Four rules, all readable at a glance:

Rule	Written as	Plain meaning
InstanceDown	<code>up == 0</code> for 1 minute	A machine has stopped answering
HighCPU	processor over 80% for 2 minutes	Something is working very hard
LowDisk	less than 15% of the disk free	The disk is filling up
LowMemory	less than 10% memory available	Memory is nearly used up

Note which ones are not about the current moment. A processor spike is not news. A processor that has been flat out for two minutes is. Good alerts describe a trend, not an instant.

Where you will meet it again. Every system that pages someone at 3am has rules like these. Writing alerts that do not cry wolf is a skill in itself, and “alert fatigue” - where people ignore alerts because there are too many - is a well-known problem.

Grafana: the chart on the wall

What it is. A program that draws Prometheus’s numbers as pictures, and lets you arrange those pictures on a page.

Analogy. The nurse has the notebook; Grafana draws the chart on the wall so everyone can see at a glance how the ward is doing.

Why we needed it. Numbers in a table are slow to read. A line going up is instant. Pictures are also how you explain a system to someone else in a meeting.

A dashboard is just a page full of charts. We used a well-known ready-made one called “Node Exporter Full” (number 1860 on grafana.com), which draws everything about a Linux machine - processors, memory, disk, network - without us designing it.

One thing we did differently from most tutorials. Rather than clicking in the web page to connect Grafana to Prometheus, we put the settings in a file in the project. So when someone downloads this project, the charts are already connected - nothing depends on a person remembering to click. **Making setup automatic instead of “here are the steps, don’t forget them” is worth more than it sounds, and interviewers notice it.**

Where you will meet it again. Grafana is extremely common for dashboards in every industry. Being able to build a useful dashboard - not a pretty one, a useful one - is a genuinely valuable skill.

Docker: shipping containers for software

What it is. A way to run a program together with everything it needs, sealed inside a standard box, so it works the same on any machine. Running several such boxes together is arranged with a file called `docker-compose.yml`.

Analogy. Shipping containers. It does not matter what is inside, the same crane can lift it and the same ship can carry it.

Why we needed it here. Prometheus and Grafana are two programs with their own requirements. Using containers meant two lines of setup instead of two software installations.

A networking detail that mattered. Containers normally live in their own little network and cannot see the virtual machines. We used a setting called **host networking** to put them on the same network as the machines, and then checked it worked before trusting it, by asking a container to fetch a page from one machine. Assumptions in networking are how afternoons disappear.

Where you will meet it again. Everywhere, and probably in your next job. The ideas to learn next after this project are how containers store data, and Kubernetes, which is the tool for running many containers across many machines.

ab: pretending to be a crowd

What it is. A small program that sends many web requests at once, to see how the system behaves under pressure. The name means ApacheBench.

Analogy. Hiring 20 people to call the restaurant at the same moment, to find out whether one waiter is enough.

Why we needed it. "Is this fast?" is not answerable by clicking a page once. You need many visitors at the same time, repeatedly, with numbers recorded.

The command, and what it means:

```
ab -n 2000 -c 20 http://192.168.122.12/labapp/work
```

- `-n 2000` means "send 2000 requests in total".
- `-c 20` means "20 of them at the same time".
- The address is the deliberately expensive page from Chapter 3, because we want the system to work hard, not idle.

Where you will meet it again. Load testing is standard practice before a big launch. `ab`'s more powerful relatives are `wrk`, `hey`, `JMeter` and `k6`.

The two numbers that matter: throughput and p95

Throughput (also “requests per second”) is how much work the system finishes per second.

Analogy. How many bowls of soup the kitchen serves per minute.

p95 latency is the time 95 out of every 100 requests took. Not the average - the slow ones.

Analogy. The average waiting time might be one minute, but if one guest in twenty waits fifteen minutes, that guest is writing an angry review. The average hides them; p95 finds them.

Why p95 instead of the average. Averages let a few terrible results hide behind lots of quick ones. Real users experience the slow ones. This is why real teams watch p95 and p99, and saying so in an interview marks you as someone who has actually looked at a dashboard rather than just read about one.

What we found, and why the answer was interesting

We measured the system with 1, then 2, then 4 processors on the kitchen machine:

Kitchen processors	requests/sec	p95	kitchen CPU	waiter CPU	whole laptop CPU
1	17.93	1497 ms	100%	14%	16%
2	29.18	971 ms	100%	14%	31%
4	44.14	761 ms	100%	14%	56%

The disappointing-looking headline: four times the processors gave only 2.46 times the soup. Not four times.

Why that is actually the good result. Because of what the numbers rule out:

- The kitchen is pinned at **100%** every single time. So the kitchen really is the thing holding everything back. Good - we found it.
- The waiter stays at **14%**. So the front door is not the problem, and 44 soups per second is not an artefact of the waiter being slow.
- The whole laptop never goes above **56%**. So we did not “run out of computer”, and the crowd-generating program was not itself the limit.
- The laptop has **4 real processor cores**, each pretending to be two. At 4 processors, the kitchen is using every real core - and sharing them with the waiter, the nurse (Prometheus), the chart (Grafana) and the crowd itself. Sharing a core with someone else slows you down, and that is why the third and fourth processor add less than the first and second.

What we genuinely could not tell apart. Whether the flattening is caused by core sharing, or by the Java program’s own internal bottlenecks (some parts of a program

simply cannot be parallelised). Both produce exactly the same shape of curve. Saying “I cannot separate those two with the data I have” is a stronger answer than inventing a reason.

Being honest about the numbers. We ran the experiment twice and got 14.71 / 23.88 / 36.65 the first time and 17.93 / 29.18 / 44.14 the second. That is a consistent 20% difference across all three points - which means something systematically changed between runs, rather than random wobble. So the shape is the finding, and the exact numbers are approximate. **Quoting 44.14 as if it were precise would be dishonest, and honesty about measurement is a professional skill, not a weakness.**

Capacity planning: how many cooks do we need?

What it is. Using measurements to decide how big a system should be, instead of guessing.

Analogy. Deciding how many tables to put in a restaurant by counting how many guests actually arrive, rather than by what feels about right.

The idea in three steps, which is really the takeaway from this whole chapter:

1. **Measure** the system as it is now. (One cook: 17.9 soups per second.)
2. **Change one thing** and measure again. (Two cooks: 29.2. Four: 44.1.)
3. **Find the limit.** Where does adding more stop helping, and what runs out first?

Why step 3 is the valuable one. Adding cooks forever is expensive, and past a point it does nothing at all. Knowing where the ceiling is - and being able to say why - is what separates “I ran a load test” from “I did capacity planning”.

Where you will meet it again. Every company that pays for computers. Cloud bills are directly tied to how big you make things, and being the person who can say “we need this size, and here is the measurement” is genuinely valuable to an employer.

6. Everything that broke, and what it teaches

This is the most valuable chapter in the guide. A project where nothing broke teaches you almost nothing, and an interviewer knows it. Every story below is real, in the order it happened.

Each one has the same five parts: what we saw, what we guessed, what it really was, how we fixed it, and the lesson.

Story 1: The bridge that would not come back

What we saw. Every time Windows' WSL restarted, the private network for our machines refused to start:

```
error: Failed to start network default
error: internal error: Network is already in use by interface virbr0
```

What we guessed. A broken configuration. Maybe a wrong setting somewhere.

What it really was. Remember the bridge from Chapter 2 - the virtual switchboard called `virbr0` that the machines plug into. When WSL restarts, it does not restart the whole kernel; it restarts the Linux user space on top of the same kernel. The switchboard lives in the kernel's part, so it survived. The program that manages it (libvirt) did not. So the new manager woke up, saw a switchboard it had not created, and refused to touch it. It was being cautious, not broken.

How we fixed it. Delete the orphaned switchboard before the manager starts:

```
sudo ip link delete virbr0
virsh net-start default
```

Then, because it happened every single time, we made it automatic with a small script that runs just before the manager starts.

The lesson. When a message says "already in use", ask who is using it and why the other thing does not know about it. The word "already" is a clue that something survived when it should not have. Also: this problem does not exist on a normal Linux server, because a real reboot destroys everything together. **Knowing that your environment has quirks, and writing them down, is part of the job.**

Story 2: The guard that could never say "no"

What we saw. Nothing. That is what makes this story dangerous.

What happened. The automatic fix from Story 1 needs a safety check. It must delete the switchboard only when no machines are running - otherwise it would rip the network out from under a working machine. So we wrote:

```
if no qemu process is running; then delete the switchboard; fi
```

And we wrote the check as `pgrep -x qemu-system-x86_64`.

What it really was. Linux shortens every program's name to **15 characters**. The program is called `qemu-system-x86_64`, which is 18. So its name is stored as `qemu-system-x86`. Our check was looking for a name that never exists, so it **never found anything**, so the answer to "is a machine running?" was always "no". Which meant the guard was not a guard at all: it would have deleted the network switchboard even with machines running, pulling the rug from under them.

How we fixed it. Ask a question whose answer cannot be wrong: does the switchboard have anything plugged into it?

```
if the switchboard has no ports attached; then delete it; fi
```

That is the actual state of the world, not a guess based on a name.

The lesson. A check that can never fail is not a check. It looks right in review and it is catastrophic in production, because it gives you the feeling of safety without the substance. Also, this is why we tested it against a live machine instead of trusting it. The most dangerous line of code in any system is the one that decides whether to do something destructive.

Story 3: It changed nothing, and it was lying

What we saw. Our proudest result was that running the setup tool twice changed nothing the second time. Then we noticed the deploy job reported a change **every** time - even when we had changed nothing at all.

What we guessed. A bug in Ansible. Or a file permission problem.

What it really was. We rebuilt the app file, and the rebuild produced a different file even though the three source files were identical. The cause: `zip` writes extra hidden information (high-precision timestamps and similar) into archives. Different bytes meant Ansible honestly reported "this file is different, I copied the new one".

How we fixed it. One extra flag tells `zip` to leave that extra information out:

```
zip -q -X -r app.war .  
#      ^ this one
```

Two rebuilds now produce identical fingerprints, and the second run of the setup tool reports `changed=0` properly.

The lesson. An idempotency claim is only as good as the thing you feed it. We had tested the tool; we had not tested whether the input was stable. If a thing that should

be identical is not identical, every clever check built on top of it is meaningless. This is why the project now prints fingerprints during builds - it turns an invisible problem into a visible one.

Story 4: The backup of nothing, which verified perfectly

What we saw. Nothing yet. This one we caught by thinking, not by failing.

The problem. Our backup tool packs a folder, writes fingerprints for everything inside, then opens the box and checks the fingerprints match. If the folder is **empty**, then the list of fingerprints is empty, the restored list is also empty, and the two empty lists are identical. The tool would announce:

```
RESTORE VERIFIED: 0 files match
```

A backup of nothing, reporting perfect health, every night, forever.

Why this was not hypothetical. During the build, our application folder really was empty for a while. A tool without this guard would have “successfully” backed up a week of nothing, and we would have found out during an emergency.

How we fixed it. Four lines at the top of the tool: if the folder has no files, refuse to run and say why.

The lesson. A checker must be able to fail, and you must have thought about what “empty” means. Most systems do not fail loudly; they fail quietly and convincingly. Whenever you write a check, ask: what input would make this check pass while being completely wrong?

Story 5: The command that printed nothing

What we saw. We had deleted the application folder on purpose (as part of the snapshot drill), but when the tool listed the folder before deleting it, the list was **empty**. That made no sense - the app had been running seconds earlier.

What we guessed. Something had lost our files. We spent several turns on a theory about data being lost during a hard shutdown, because the machines had been restarting unexpectedly at the time.

What it really was. Boring. The connection to the machine dropped mid-command, and we had told the command to throw away its error messages:

```
ssh ... "ls /var/lib/tomcat10/webapps/" 2>/dev/null  
#                ^^^^^^^^ this
```

The error message went into the bin. All we saw was silence - and **silence looks exactly like an empty folder**.

How we fixed it. Stop hiding errors on the commands that are supposed to prove something. And the real fix was mental: after that, every destructive step in the project

was bracketed by fingerprint checks, so “the file is missing” could never be confused with “the command did not run”.

The lesson. Never silence the error output of the command you are using as evidence. The first theory had been wrong, and it cost hours. The fix was not clever: it was to make the evidence unambiguous. Two rules worth keeping forever: do not throw away error messages, and never let “no output” count as proof.

Story 6: The machines kept rebooting themselves

What we saw. The virtual machines rebooted every few minutes. Our work kept vanishing. Commands failed with “no route to host”. It looked like the lab was falling apart.

What we guessed, and ruled out one by one. This is worth reading as a method, because the guessing is what took the time:

- The idle timer. WSL switches itself off when nothing is happening. We turned that off and told it never to do it. **Not the cause.**
- The memory killer. Linux has a service that kills programs when memory runs out (`systemd-oomd`). Check whether it killed our virtual machines. **It was not even installed.**
- The watchdog. Each virtual machine has a “dead man’s handle” device that reboots it if it stops responding. If it were armed, it would reboot under load. We checked whether anything was arming it. **Nothing was.**
- The manager crashing. We looked at how many times libvirt had crashed. **Zero times.**

What it really was. We finally looked at the kernel’s own log and saw the same boot messages repeating every few minutes:

```
Linux version 6.18.40.1-microsoft-standard-WSL2 ...  
WSL (1 - mini_init): WSL user cgroup created ...
```

but at the same time, the machine’s **uptime** and its **boot identity** said “nothing has restarted”. Both facts were true, and together they were the answer: WSL keeps its kernel running, but **tears down the Linux user space as soon as the last session closes**. Closing a session kills the virtual machine manager, which takes the virtual machines down with it. Our machines only came back because we had told them to start automatically.

So the “reboots” were not crashes at all. They were the lab being switched off every time we stopped typing, and switched back on when we typed again.

How we fixed it. Keep a session open so WSL never decides to tear things down:

```
nohup wsl.exe -e bash -c "sleep 3600" >/dev/null 2>&1 &
```

Then we verified it: we watched the process start times and the manager’s start time, waited 90 seconds doing nothing, and looked again. Unchanged. Fixed.

The lesson. “The tool said it restarted” and “the machine restarted” are different claims. The breakthrough came from noticing two facts that could not both be

true, and treating that contradiction as the clue instead of ignoring it. And the method note matters too: we ruled out four plausible causes before finding the real one, which is normal. Debugging is mostly the discipline to eliminate without getting bored.

Story 7: The processor arrived, but stayed asleep

What we saw. We added processors to a running machine. The manager reported success: “2 processors”. The machine itself said “I have 1”.

What we guessed. The change had not really happened.

What it really was. The machine’s log held the answer:

```
ACPI: CPU1 has been hot-added
```

The processor had been delivered and plugged in - and left **switched off**. The command we were using to count processors counts the ones that are awake. One extra command to switch it on, and the count was correct.

Why nobody documented it. Most guides check “did the tool report success?” and stop. Success at the top does not guarantee success all the way down.

How we fixed it. Switch the processor on explicitly before measuring, and document that the obvious verification step is not enough.

The lesson. Verify at the place the work is used, not where it is issued. The manager’s opinion is not evidence. The machine’s own report is.

Story 8: The clock that went backwards

What we saw. After we rewound a machine to a snapshot, its clock jumped backwards. The machine noticed and said so in its log. About a minute later, it restarted on its own.

What we guessed. Something was badly wrong - possibly the watchdog from Story 6, which would have been genuinely serious, because a watchdog firing during a load test would silently ruin all our measurements.

What it really was. Rewinding to a snapshot restores a moment, including the clock reading from that moment. And it invalidates anything the machine was in the middle of doing at that moment, such as a write to disk. The machine notices both, and can decide to restart to be safe. Not a fault - a side effect of time travel.

How we fixed it. Nothing to fix, but we checked the scary possibility properly: it was not the watchdog. That mattered, because if it had been armed, it could have restarted a machine in the middle of measuring, and we would have published wrong numbers without knowing.

The lesson. When you use time travel, expect the clock and unfinished work to be wrong. And: distinguish “harmless side effect” from “the same symptom caused by something that would ruin my results”. We spent a few minutes on the second question and it was the right few minutes.

The five lessons worth memorising

If you remember nothing else from this chapter:

1. **A check that can never fail is not a check.** Test your safety net by trying to break it, with a live system.
2. **A claim is only as good as the thing you are claiming about.** If a thing that should be identical is not, everything built on it is meaningless.
3. **Never silence errors on a command that is supposed to prove something.** “No output” is not evidence of anything.
4. **The tool saying it worked is not the same as it working.** Verify where the work is used.
5. **Write it down.** Every story here is in the build log with the commands and the output. A problem you solved and did not record is a problem you will solve twice.

7. Glossary, cheat sheet, and what to learn next

Glossary, in plain words

Ansible - A program that reads a written checklist and makes machines match it. The manager with the clipboard.

Apache - The waiter. The web server that greets requests first and passes them to the kitchen.

ab - ApacheBench. A program that pretends to be many visitors at once, to see how a system behaves under pressure.

backing file - The big read-only disk that thin disks are layered on top of, like tracing paper over a drawing.

backup - An independent copy kept somewhere else, which is checked. The copy of the recipe at grandma's house.

bridge (virbr0) - The virtual switchboard that virtual machines plug into.

capacity planning - Deciding how big a system should be by measuring it, rather than guessing.

checksum / fingerprint (SHA-256) - A short string computed from a file. Change one letter and it changes completely. Used to answer "is this the same as that?"

cloud image - A finished, pre-installed operating system in a file. A frozen meal instead of cooking from scratch.

cloud-init - The note you leave for a brand-new machine telling it who it is and how to let you in.

commit - One saved step in the history of a project, with a message about why.

container (Docker) - A program sealed together with everything it needs, so it runs the same anywhere. A shipping container.

cron / timer - The alarm clock that runs jobs on a schedule.

DHCP - The automatic address giver on a network.

DHCP reservation - "This machine always gets this address", matched by the machine's hardware serial number.

emulation - Pretending to be a different processor, in slow motion. What happens when hardware virtualization is unavailable.

firewall (ufw) - The bouncer with the guest list. Decides which doors are open to whom.

Grafana - The chart on the wall. Draws monitoring numbers as pictures.

guest - A virtual machine, from the point of view of the host.

handler - An Ansible job that only runs if something actually changed. "Restart the service, but only if you edited its settings."

host - The real machine that runs virtual machines; or, the thing a container runs on.

hot-add / live resize - Adding memory or processors to a machine that is switched on.

hypervisor - The boss program that creates and manages virtual machines. KVM, VMware, Hyper-V.

idempotent - Doing it twice has the same result as doing it once. A light switch, not a doorbell.

inventory - Ansible's address book of machines.

JAR / WAR - The sealed box containing a Java program (`.jar`) or a whole Java web application (`.war`).

Java / JVM - The language and the program that runs it. The Esperanto of programming languages.

JSP - A web page with small pieces of Java code inside it.

KVM - The virtualization feature built into Linux. The engine.

libvirt - The manager that remembers virtual machines and gives you tidy commands for them. Talked to with `virsh`.

manifest - A list of the fingerprints of every file in a backup. The inventory list taped to the box.

metric - A number with a timestamp. A row in the nurse's notebook.

NAT - A private network that can reach out to the internet, while the internet cannot reach in.

node_exporter - The little meter on each machine that reports how busy it is.

p95 latency - The time 95 out of 100 requests took. Finds the slow ones that an average would hide.

playbook - The whole checklist in Ansible (`site.yml`).

Prometheus - The nurse with the notebook. Collects and stores monitoring numbers, and checks alert rules.

PromQL - The little language for asking questions of monitoring numbers.

pull request (PR) - A polite request to add your work to the main version, with a chance to look at it first.

qcow2 - The disk file format that only uses space for data you actually write, and can be layered on a base image.

qemu - The program that pretends to be the hardware of a virtual machine. The car body and wheels.

reverse proxy - A rule that quietly hands requests to another machine and brings the answers back. A phone answering service.

role - A group of Ansible jobs for one purpose, such as `tomcat`.

rsync - The careful mover that copies only what changed and can be interrupted.

scrape - Prometheus fetching numbers from a machine on a timer.

snapshot - A photograph of a machine at one moment, stored on the same disk. For fast undo, not for disasters.

SSH key - A padlock you hand out and the only key that opens it. Replaces passwords.

systemd - The part of Linux that starts and supervises programs.

task - One line on the Ansible checklist.

throughput - How much work is finished per second. Bowls of soup per minute.

Tomcat - The kitchen. A Java web application server.

virtual host - A section of a web server's settings for one website.

virtualization - Making a computer inside a computer. A folded paper house inside a bigger room.

virsh - The command you type to talk to libvirt. Talking to the garage manager.

WSL2 - Microsoft's way of running real Linux inside Windows. A guest flat built into your house.

Command cheat sheet

Machines

```
virsh list --all           # list machines and whether they are running

virsh dominfo app01        # full detail: processors, memory, state
virsh start app01 / shutdown app01 # switch on / ask politely to switch off
virsh destroy app01        # pull the plug (use with care)
virsh console app01        # look at its screen (leave with Ctrl + J)

virsh net-list --all       # is the private network up?
virsh net-dhcp-leases default # which machine has which address
virsh domblklist app01     # which disk file is it using
virsh autostart app01      # start it automatically when the host boots
```

Snapshots and resizing

```
virsh snapshot-create-as app01 my-photo --description "... "
virsh snapshot-list app01
virsh snapshot-revert app01 my-photo
virsh setvcpus app01 2 --live      # add processors while running
virsh setmem app01 3G --live       # add memory while running
# inside the machine, switch a new processor on:
echo 1 | sudo tee /sys/devices/system/cpu/cpu1/online
```

Automation

```
ansible all -m ping           # can I reach every machine?
ansible-playbook site.yml      # make the machines match the checklist
ansible-playbook site.yml --limit app # only the kitchen machine
```

The application

```

bash app/build.sh # build the WAR
curl http://192.168.122.12/labapp/health # through the waiter
curl http://192.168.122.11:8080/labapp/health # straight to the kitchen
(blocked)
ssh -i ~/.ssh/kvmlab ops@192.168.122.11 # log in to the kitchen

sudo journalctl -u tomcat10 -n 50 --no-pager # the kitchen's diary

```

Backups

```

sudo systemctl start tomcat-backup.service # make a backup now
sudo journalctl -u tomcat-backup.service -n 20 # did it verify?
systemctl list-timers tomcat-backup.timer # when does it run next?

sudo /usr/local/bin/tomcat-backup.sh verify <archive> # check one by hand
bash scripts/pull-backups.sh # copy them off the machine

bash scripts/backup-failure-drill.sh # prove the checker works

```

Monitoring

```

cd monitoring && docker compose up -d # start Prometheus + Grafana

docker compose logs --tail 20 # what are they saying?

curl -s localhost:9090/api/v1/targets # are the machines reporting?

curl -s localhost:3000/api/health # is Grafana alive?

```

Measuring and evidence

```

bash scripts/capacity-test.sh # the load test sweep
bash scripts/capture-evidence.sh # regenerate the evidence

ab -n 2000 -c 20 http://192.168.122.12/labapp/work # one load test by hand

```

The lab's two house rules

```

# 1. Before a session, keep a WSL session open so the machines are not
#    switched off between commands:
nohup wsl.exe -e bash -c "sleep 3600" >/dev/null 2>&1 &

# 2. After any WSL restart, check the network came back:
virsh net-list --all && ping -c1 192.168.122.1

```

Where you will meet all of this again

If the job says...	This project covers...
VMware / Hyper-V / KVM	Chapters 2 and 4: machines, snapshots, hot-add, the mapping table
Linux administration	Every chapter: systemd, users, permissions, logs, SSH
Ansible / configuration management	Chapter 4: roles, idempotency, and proof of it
Tomcat / WebLogic / Apache / nginx	Chapter 3: app servers, reverse proxies, virtual hosts
Shell scripting	The backup tool, the VM builder, the load test - all shell
Backup, DR, restore testing	Chapter 4 and the disaster recovery drill
Monitoring, Prometheus, Grafana	Chapter 5
Capacity planning	Chapter 5: the whole experiment
Networking, firewalls	Chapters 2 and 3
Containers, Docker	Chapter 5
Git, code review, CI	Chapter 4: branches, pull requests, clean history

Sentences worth being able to say out loud

Short versions you can say in an interview without rambling:

- **Idempotency:** “Running the playbook twice reports `changed=0` on the second run, which means every task inspected the machine and found nothing to do. To keep that true I had to make sure my build produced identical files, because a rebuild that changes bytes makes the deploy task report a change forever.”
- **Snapshot versus backup:** “A snapshot lives on the same disk as the machine, so it is for fast rollback. A backup is an independent, verified copy elsewhere, for disaster recovery. I proved the difference by destroying the machine - the snapshot died with the disk, and the backup brought the application back.”
- **Backup verification:** “I restore the archive into a scratch directory and compare checksums against a manifest from the source, so I am verifying the restore rather than trusting the archive. I also tamper with a manifest and confirm the checker refuses it, because a checker that only ever passes is worthless.”
- **Capacity:** “I load-tested through the proxy and read CPU back from Prometheus. The app VM was pinned at 100% at every size, the proxy sat at 14% and the host never passed 56%, so the app was the bottleneck and neither the proxy tier nor host exhaustion explained the flattening.”
- **Honesty about measurement:** “The same experiment run twice differed by about 20%, which is systematic rather than noise, so I quote the shape of the result rather than the exact numbers.”

- **Something that broke:** “My safety guard used `pgrep -x qemu-system-x86_64`, which can never match because Linux truncates process names to 15 characters. So the guard was always true and would have deleted the network bridge out from under running machines. It looked correct in review; only testing it against a live machine caught it.”

What to learn next, in a sensible order

1. **The same lab, automated one level higher.** Terraform can create the virtual machines themselves, instead of a shell script. Same idea as Ansible but for machines rather than settings inside machines.
2. **Automatic checks on every change.** GitHub Actions running `ansible-lint` and `yamllint` when you push. That is the beginning of CI.
3. **A second kitchen and a load balancer.** Add a second app machine and let Apache share orders between them, then switch one off and show that no requests fail. That is the beginning of high availability.
4. **HTTPS.** A self-signed certificate on Apache. It teaches how TLS actually works, and it is the first thing a security reviewer asks about.
5. **Containerise the application itself.** Build a Docker image that contains Tomcat and the WAR, so the app can be started anywhere in one command.
6. **Kubernetes.** The moment you have more than a couple of containers, people reach for it. It is a big subject; learn it after you have done steps 1 to 5.
7. **Something with a database.** Add a real database, and the backup conversation becomes genuinely interesting, because a database cannot just be copied while it is running.

The one-page summary

You built two computers inside your computer. One is a waiter that greets visitors, one is a kitchen that makes the food. You wrote a checklist so a program could build both machines identically, and proved it was safe to run twice. You took a photograph so you could undo mistakes in seconds, and you kept a copy of the recipe somewhere else so you could survive a disaster - then proved it by destroying the kitchen entirely and rebuilding it in under six minutes. You put meters on both machines, drew the numbers on a wall, and used them to find out exactly which part was the slow one. Then you locked the doors so visitors could only come through the front.

The part that will get you hired is not that it worked. It is that you can say what broke, why it broke, how you found out, and what you would do differently - because that is the job.

Part two: the README

kvm-tomcat-lab

A two-tier Java application tier built on KVM virtual machines: Apache reverse-proxies to Tomcat 10. Provisioned with idempotent Ansible, protected by snapshots and a checksum-verified backup, monitored with Prometheus and Grafana, and sized with a measured load test rather than a guess.

Everything below is measured output from this lab. The raw output behind each claim is in [docs/evidence/](#) and can be regenerated with `bash scripts/capture-evidence.sh`.

Just want to understand it? There is a 42-page plain-English study guide that explains every tool in this project - what it is, why it was needed, how it is used here, and where you will meet it in a real job - starting from zero assumed knowledge. Read the PDF: [kvm-tomcat-lab-study-guide.pdf](#), or the Markdown sources in [docs/study-guide/](#) (rebuild with `bash docs/study-guide/build-pdf.sh`).

Architecture

```
Windows 11 host - Intel i5-1155G7, 4 cores / 8 threads, 20 GB RAM
├─ WSL2 Ubuntu 24.04 (nested virtualisation, /dev/kvm available)
│   └─ KVM / libvirt
│       └─ NAT network 192.168.122.0/24
│           ├── web01 192.168.122.12  Apache 2.4      1 vCPU / 1 GB
│           └── app01 192.168.122.11  Tomcat 10      1-4 vCPU / 2-4 GB
│                                   labapp.war    maxvcpus 4 set at creation
└─ Prometheus 3.15.0 + Grafana 13.2.3 (host networking, scrapes :9100)
```

Request path: client → `web01:80` (Apache) → `app01:8080` (Tomcat) → `labapp`. Only `web01:80` is reachable from outside the lab; `app01:8080` is firewalled to the web tier, which is what makes the reverse proxy load-bearing rather than decorative.

`app01` is deliberately built undersized at 1 vCPU. That is not an oversight - it exists so the load test has a bottleneck to find and a scaling story to tell.

Measured results

Claim	Result	Evidence
Ansible is idempotent	second run: <code>app01 changed=0</code> , <code>web01 changed=0</code>	01
Apache proxies to Tomcat	<code>curl</code> through <code>web01</code> → <code>OK</code> <code>app01</code>	02
Tomcat is not directly reachable	<code>app01:8080</code> from the host: HTTP 000, <code>curl</code> exit 28 (dropped)	02
Backup restores, not just archives	RESTORE VERIFIED: 5 files match	03
Backup verification rejects a bad backup	tampered manifest → RESTORE MISMATCH, exit 1	04
Backup exists off the VM	2 archives in <code>~/kvm-lab-backups</code> , <code>sha256sum -c</code> OK	05
Snapshot rollback	5.1 s revert, no reboot, content hash unchanged	snapshots
Live resize	memory hot-added live; a hot- added vCPU stays offline until online	07
Monitoring	2 targets <code>up</code> , 4 alert rules, dashboard imported by script	08
Recovery from total VM loss	348 s (5 min 48 s) to a working service	disaster-recovery

Snapshots (KVM console, `virsh list` and `dominfo`) - see [docs/screenshots/](#).

Capacity planning

`bash scripts/capacity-test.sh` hot-adds vCPUs, online them inside the guest, and runs `ab -n 2000 -c 20` against `/labapp/work` at each size. Peak CPU is read back from Prometheus rather than eyeballed off a graph.

app01 vCPUs	guest <code>aproc</code>	req/sec	p95	app01 CPU	web01 CPU	host CPU
1	1	17.93	1497 ms	100%	14%	16%
2	2	29.18	971 ms	100%	14%	31%
4	4	44.14	761 ms	100%	14%	56%

4x the CPUs bought **2.46x** the throughput, and p95 improved **2.0x**.

What the table rules out, which is the more useful half of the result:

- **app01 is the bottleneck.** Its CPU is pinned at 100% at every size, so the application really is CPU-bound.
- **The proxy tier is not the constraint.** web01 sits at 14% regardless.
- **The host is not exhausted**, rising 16% → 56% and never saturating, so the flattening is not “out of machine” and the load generator is not the limit.
- **The host is 4 physical cores with SMT.** At 4 vCPUs, app01 occupies every physical core, sharing them with the load generator, Apache, Prometheus and Grafana. Sibling hyperthreads share execution units, which explains why the 3rd and 4th vCPU add less than the 1st and 2nd.

What the data cannot separate: shared-core contention from the JVM’s own serial parts (GC, single-threaded components). Both produce the same curve, and telling them apart needs the JMX exporter.

Measurement honesty: the same experiment run twice differed by a consistent ~20% per data point (14.71 / 23.88 / 36.65 once, 17.93 / 29.18 / 44.14 again), which is systematic rather than noise. The shape reproduced and is the finding; the absolute requests per second are indicative, not precise. Full analysis in [capacity-planning.md](#).

Reproduce

Requires a host with virtualization support. Steps 1-3 take about 20 minutes.

```
# 1. KVM host: qemu, libvirt, virtinst, cloud image. See docs/host-setup.md
sudo apt-get install -y qemu-system-x86 qemu-utils libvirt-daemon-system \
    libvirt-clients virtinst cpu-checker zip
sudo systemctl enable --now libvirtd

# 2. Two VMs with fixed addresses, from the cloud image
bash scripts/create-vm.sh

# 3. Build the WAR and configure both VMs
bash app/build.sh
ansible-playbook site.yml

# 4. The application answers, through the proxy
curl http://192.168.122.12/labapp/health
# OK app01
```

Optional extras:

```
cd monitoring && docker compose up -d          # Prometheus :9090, Grafana :3000
bash monitoring/import-dashboard.sh             # Node Exporter Full (ID 1860)
bash scripts/capture-evidence.sh               # regenerate docs/evidence/
bash scripts/capacity-test.sh                 # the capacity table above
bash scripts/backup-failure-drill.sh           # prove the backup checker fails
```

Snapshot versus backup

Interviewers ask this every time.

A **snapshot** is a point-in-time state that lives on the same storage as the VM. It is for fast rollback before a risky change. It does not protect you from losing the disk - which this project demonstrates literally: the disaster recovery drill destroys `app01` with `--remove-all-storage`, and the `clean-deploy` snapshot dies with it.

A **backup** is an independent copy stored elsewhere and verified. It is for disaster recovery. Here that means an archive of the deployment, a SHA-256 manifest, a restore-to-scratch comparison, and a copy pulled off the VM whose checksum is re-checked after the transfer.

How this maps to VMware and Hyper-V

The lab used KVM. These are the equivalents, and being clear about which tooling was actually used is stronger than implying experience that does not exist.

KVM/libvirt (used here)	vSphere	Hyper-V
<code>virsh snapshot-create-as</code>	VM snapshot	Checkpoint
<code>virsh snapshot-revert</code>	Revert to snapshot	Apply checkpoint
<code>virsh setvcpus/setmem --live</code>	Hot-add CPU/memory	Dynamic Memory, processor change
qcow2 backing-file overlay	Linked clone	Differencing disk
<code>maxvcpus</code> / <code>maxmemory</code> at creation	VM limits fixed at creation	Startup/maximum RAM
<code>libvirt default</code> NAT network	Standard vSwitch / port group	Virtual switch (NAT/internal)
<code>virsh migrate</code>	vMotion	Live Migration

Lessons learned

Five real things broke. The full list, including the ones that cost hours, is in [the build log](#).

1. A guard that can never be false is not a guard. A WSL restart leaves libvirt's `virbr0` bridge behind, which blocks the network coming up, so a script deletes the stale bridge before `libvirtd` starts - guarded by `pgrep -x qemu-system-x86_64`. Linux truncates process names to 15 characters, so that pattern matches nothing, the negation is always true, and the “safety” check would have deleted the network out from under running VMs. Replaced with a check of the bridge's own attached ports. It looked correct in review; only testing it against a live VM caught it.

2. An idempotency claim is only as good as the artifact. `zip` writes extra metadata, so rebuilding an unchanged WAR produced a different file, the Ansible copy task reported `changed` forever, and the `changed=0` proof was quietly false. `zip -X` fixed it. The lesson generalises: verify that the thing you feed into an idempotency check is itself reproducible.

3. WSL tears down the distro when the last session exits. That stops `libvirtd`, which takes `qemu` with it, so the guests were power-cycled between almost every

command and my work kept vanishing. Found by noticing that kernel boot messages repeated while `boot_id` and `/proc/uptime` carried on unchanged - the VM kernel stays alive, only the distro is torn down. Fixed by holding a session open, and verified by watching process start times stop changing.

4. An empty directory archives and verifies perfectly. Two empty manifests compare equal, so a backup tool without a guard prints `RESTORE VERIFIED: 0 files match` for a deployment directory containing nothing. The deployment really was empty for a while during this build, so the guard against it is in the tool.

5. Never swallow stderr on the command that is supposed to prove something. A destructive command had worked, but the listing before it printed nothing because the SSH connection dropped and the error went to `/dev/null`. An empty output looked exactly like an empty directory, and I spent several turns chasing a lost-write theory that did not exist.

Limitations

Stated plainly, because a portfolio project that lists no weaknesses is not credible:

- **Single host, no high availability.** One app VM. If `app01` is down, the site is down; the firewall is not redundancy.
- **Absolute performance numbers describe a laptop.** Nested virtualisation on an i5-1155G7 with 4 physical cores. The method and the shape of the scaling curve are the transferable parts.
- **The off-VM backup copy is on the same physical machine.** It survives losing the VM, not losing the host.
- **The backup covers the deployment, not the machine.** In this lab the WAR is also in git, so the backup is not the only recovery path - a truer test would involve state that git does not hold.
- **SSH is open to any address** and there is no TLS on port 80. Both are deliberate so the lab stays reachable; [hardening.md](#) lists these and the other gaps.
- **Lab-grade settings** that would not ship: host key checking disabled in `ansible.cfg`, Grafana on `admin/admin`, no secrets management.
- **Not tested on VMware or Hyper-V.** The concepts map (see the table above), but the tooling used was KVM.

Repository layout

```
.
├── ansible.cfg, inventory.ini, site.yml
├── app/                    health.jsp, work.jsp, web.xml, build.sh -> labapp.war
├── roles/
│   ├── common/           apt cache, curl, htop, rsync
│   ├── tomcat/           Tomcat 10, deploys the WAR, health-checked
│   ├── apache/           reverse proxy vhost
│   ├── backup/           verified backup script, systemd service + timer
│   ├── node_exporter/    metrics agent
│   └── hardening/        ufw rules matching the two-tier design
├── monitoring/           Prometheus, Grafana, alert rules, dashboard import
├── scripts/              create-vm, capacity-test, evidence, DR helpers
└── docs/
    ├── kvm-tomcat-lab-study-guide.pdf  plain-English guide to every tool used
    ├── study-guide/                   the study guide sources, one file per chapter
    ├── build-log.md                   module-by-module log, including what broke
    ├── host-setup.md                 KVM host install and the WSL quirks
    ├── snapshots-and-resize.md
    ├── backup-and-restore.md
    ├── capacity-planning.md
    ├── hardening.md
    ├── disaster-recovery.md
    ├── interview-notes.md            the questions this project prepares you for
    └── evidence/                    raw output behind every claim in this README
```

Definition of done

- Both VMs run and are reachable over key-only SSH
- `ansible-playbook site.yml` twice, second run `changed=0`
- `curl` via Apache returns `OK app01`
- Snapshot taken, deliberate breakage, revert works (5.1 s)
- Live vCPU/memory resize shown with `nproc` and `free -m`
- Backup shows `RESTORE VERIFIED`; tamper drill shows `RESTORE MISMATCH`
- Backup copied to the host and checksum-verified
- Prometheus targets UP, Grafana dashboard working
- Load test table filled in with real numbers
- Disaster recovery drill: total VM loss to working service in 348 s
- README complete with a real “Lessons learned”
- Study guide written for someone starting from zero (42 pages)

Stack

KVM / libvirt, cloud-init, qcow2, Ansible, Apache 2.4, Tomcat 10 (Jakarta EE), JSP, systemd timers, tar/SHA-256 integrity verification, rsync, Prometheus, Grafana, ufw, bash.

Built on Ubuntu 24.04 LTS.

Part three: technical write-ups

KVM host setup

Module M1. The “host” for this lab is a **WSL2 Ubuntu 24.04 distro** on Windows 11 Pro using nested virtualization, not a bare-metal Ubuntu box. That is the weakest of the three options in the project guide (bare metal > Ubuntu VM with nested virt > WSL2), so it is worth writing down what that costs.

Environment

Thing	Value
Windows host	Windows 11 Pro, 19.8 GB RAM
WSL distro	Ubuntu 24.04.3 LTS (noble), systemd as PID 1
CPUs visible to WSL	8
RAM visible to WSL	~9.6 GB
Free disk	~950 GB
<code>/dev/kvm</code>	present, crw-rw---- root kvm
Repo path	<code>~/kvm-tomcat-lab</code> (native ext4)

Why the repo is not on `/mnt/c`

The repo lives in the WSL filesystem, not in the Windows project folder.

- `git` over the 9p/drvfs bridge to `/mnt/c` is slow and mangles file modes.
- Windows-side OneDrive sync fighting `.git` corrupts repositories.
- `ansible-playbook` and `qemu-img` want a normal POSIX filesystem.

Windows can still read it at `\\wsl.localhost\Ubuntu\home\ziaur26261\kvm-tomcat-lab`.

Why WSL2 nested virtualization

Nested virt means the KVM guests run inside the WSL2 Hyper-V VM. It works because Windows exposes virtualization extensions to WSL2 (on by default on current Windows 11 builds), and the WSL2 kernel ships the KVM modules.

Costs, stated honestly:

- guest CPU is slower than bare metal, so the M7 capacity numbers are relative, not absolute;
- total RAM available to the guests is capped at roughly 9.6 GB, because WSL2 defaults to 50% of physical memory;
- a WSL restart does not clean up libvirt’s bridge interface (see the quirk below).

Packages installed

```
sudo apt-get install -y qemu-system-x86 qemu-utils \
  libvirt-daemon-system libvirt-clients virtinst cpu-checker zip
sudo systemctl enable --now libvirtd
sudo usermod -aG libvirt,kvm "$USER"      # then restart the distro
```

zip is needed by `app/build.sh`. `cpu-checker` provides `kvm-ok`.

Group membership only applies to a new login session, so the distro was restarted with `wsl.exe --terminate Ubuntu` — the WSL equivalent of logging out and back in. After that `virsh` works without `sudo`:

```
uid=1000(ziaur26261) groups=...,111(libvirt),993(kvm)
```

Validation

```
$ sudo kvm-ok
INFO: /dev/kvm exists
KVM acceleration can be used

$ systemctl is-active libvirtd
active

$ virsh net-list
Name      State   Autostart   Persistent
-----
default   active  yes         yes

$ ping -c 2 192.168.122.1
2 packets transmitted, 2 received, 0% packet loss
```

The `default` libvirt network is a NAT network on `192.168.122.0/24` with the host as gateway `192.168.122.1`. Guests reach the internet and each other through it; the host reaches guests directly.

Fixed addresses

DHCP reservations by MAC, created by `scripts/create-vm.sh`. Fixed addresses mean the Ansible inventory never changes.

VM	IP	MAC	Role	Shape
app01	192.168.122.11	52:54:00:aa:00:11	Tomcat 10	1 vCPU / 2 GB, max 4 vCPU / 4 GB
web01	192.168.122.12	52:54:00:aa:00:12	Apache 2.4	1 vCPU / 1 GB, no headroom

Each VM's 10 GB disk is a qcow2 overlay on the shared 597 MB base image, so the base is stored once. `qemu-img info` shows the `backing file:` line.

Access is key-only as the `ops` user (`sudo` with no password, so Ansible can use `become`), using the lab key `~/.ssh/kvmlab`. Password auth is disabled in cloud-init.

Cloud-init

No installer runs. `virt-install --import` boots the cloud image directly and feeds it a seed ISO with a `#cloud-config` that sets the hostname, creates `ops`, and installs the SSH public key. First boot completes in seconds.

Cloud image

```
cd /var/lib/libvirt/images
sudo wget -O noble-server-cloudimg-amd64.img \
  https://cloud-images.ubuntu.com/noble/current/noble-server-cloudimg-amd64.img

sudo wget -O SHA256SUMS \
  https://cloud-images.ubuntu.com/noble/current/SHA256SUMS
sudo grep 'noble-server-cloudimg-amd64.img$' SHA256SUMS | sudo sha256sum -c -
```

```
noble-server-cloudimg-amd64.img: OK
```

597 MB, checksum verified against the publisher's `SHA256SUMS`. An unverified base image is the kind of thing that is only noticed months later.

WSL quirk: stale `virbr0` blocks the network after a restart

Restarting the distro (`wsl --terminate`, `wsl --shutdown`, or Windows sleep) leaves the `virbr0` bridge behind, because WSL keeps the VM kernel running while only the distro's init is restarted. The new `libvirtd` then refuses to bring up the network:

```
error: Failed to start network default
error: internal error: Network is already in use by interface virbr0
```

`virsh net-start default` will not fix it. Delete the orphaned bridge first:

```
sudo ip link delete virbr0
virsh net-start default
```

Permanent fix

`scripts/libvirt-clear-stale-bridge.sh` is installed as an `ExecStartPre` for `libvirtd.service`, so `virbr0` is cleared automatically before the daemon starts:

```
sudo install -m 0755 scripts/libvirt-clear-stale-bridge.sh /usr/local/bin/
sudo mkdir -p /etc/systemd/system/libvirtd.service.d
printf '%s\n' '[Service]' \
  'ExecStartPre=/usr/local/bin/libvirt-clear-stale-bridge.sh' \
  | sudo tee /etc/systemd/system/libvirtd.service.d/10-wsl-stale-bridge.conf
sudo systemctl daemon-reload
```

The guard matters. The script deletes the bridge only when nothing is attached to it, checking `/sys/class/net/virbr0/brif`. If any interface is a bridge port, a domain is live and the bridge is left alone. An unconditional delete would rip the network out from under running VMs.

An earlier version of the guard used `pgrep -x qemu-system-x86_64`, which can never match anything: Linux truncates process names to 15 characters, so the process is `qemu-system-x86` and the negation was always true. That guard looked correct, reviewed correctly, and would have deleted the bridge on every dependency restart. Testing it against a running VM is what caught it:

```
$ pgrep -x qemu-system-x86_64 ; echo $?
1                                <-- never matches, so "! pgrep" was always true
$ ls -A /sys/class/net/virbr0/brif
vnet0 vnet1                      <-- real state: two live domains attached
```

WSL idle shutdown

WSL shuts down its utility VM when idle, which hard-kills `qemu` and power-cycles the guests. They autostart again, but an unflushed guest write can be lost, and an SSH session can die mid-command. Disabled in `C:\Users\ziaur\.wslconfig`:

```
[wsl2]
vmIdleTimeout=-1
```

After that change WSL survived a 150 second idle window with no WSL processes running (uptime 25s before, 184s after) and both domains stayed up.

This is a WSL-only artifact. On a real Ubuntu host, rebooting destroys `virbr0` with the rest of the network namespace, so neither problem exists.

Verify the network self-heals after a distro restart:

```
virsh net-list --all && ping -c1 192.168.122.1
```

Snapshots, rollback and live resizing

Module M5. Everything below is measured output from this lab, not theory.

Snapshot versus backup

Interviewers ask this every time, so here is the answer this project can defend.

	Snapshot	Backup
Where it lives	Same storage as the VM (<code>/var/lib/libvirt/images</code>)	Somewhere else entirely
Good for	Fast rollback before a risky change	Disaster recovery
Survives losing the VM disk	No - it dies with the disk	Yes, that is the point
Includes RAM	Optionally, as done here	No, it is a file archive
Verified	Not applicable	Restored to scratch and checksum-compared (M6)

A snapshot is not a backup. A backup is not a snapshot. This lab has both, and M6 proves the backup restores.

Snapshot and rollback drill

```
virsh snapshot-list app01
virsh snapshot-create-as app01 clean-deploy \
  --description "Tomcat 10 + labapp deployed, WAR sha256 9e6fe29b..."
```

The project guide warns that `snapshot-create-as` can fail on a cloud image because of the cloud-init seed device. It did not fail here, because the seed ISO is already detached after first boot - `virsh domblklist app01` shows `sda` as an empty cdrom, so only the `vda` `qcow2` is involved.

The WAR checksum was recorded at snapshot time, then the deployment was destroyed deliberately:

```
$ ssh ops@192.168.122.11 'sudo rm -rf /var/lib/tomcat10/webapps/*'

through Apache:  HTTP 404
direct to Tomcat: HTTP 404
```

404 rather than 503 is the correct symptom: Tomcat is healthy, the `/labapp` context no longer exists. (A 503 would have meant Tomcat itself was down - worth knowing, because it is how you tell “app broken” from “app server broken” at 3am.)

Then the revert:

```
$ time virsh snapshot-revert app01 clean-deploy
Domain snapshot clean-deploy reverted
real 0m5.119s
```

The revert did not reboot the machine. Guest uptime was `up 1 minute` immediately before the revert and `up 1 minute` immediately after it, and the WAR came back with the same sha256 as before the damage:

```
guest WAR sha256: 9e6fe29bc4f1cf2a0739b84e6c528a35ba34880b60fe5874670e0e1fa340769a
local WAR sha256: 9e6fe29bc4f1cf2a0739b84e6c528a35ba34880b60fe5874670e0e1fa340769a
```

That is a memory-state rollback (`loadvm`), not a disk restore plus boot. The domain resumed at the exact instant the snapshot was taken, which is why recovery took five seconds instead of the two to three minutes a guest boot costs in this environment.

One client-visible wrinkle: the first request after the revert returned a `502 Proxy Error`, and the next five returned `200 OK`. The restored JVM and Apache's pooled upstream connection need a moment to agree. A rollback is near-instant but not atomic from the client's point of view - worth saying out loud rather than claiming a clean cutover.

Second wrinkle, and the more interesting one. About a minute after the revert the guest rebooted on its own, with

`systemd-resolved: Clock change detected. Flushing caches.` as the last line before the cut. Restoring a running memory image onto a disk image rewinds the guest clock and invalidates whatever I/O was in flight at snapshot time, so the guest notices something is wrong.

Worth ruling out explicitly: it is not the watchdog. The domain has `<watchdog model='itco' action='reset'/>` because that is a virt-install default, but the guest never arms it - `systemctl show -p RuntimeWatchdogUsec` returns `0`, and nothing sets it in `systemd/system.conf`. An armed watchdog would have been a much nastier bug, because it would fire during the M7 load test and silently invalidate the capacity numbers.

Practical consequence: the checksum proof still holds (the WAR came back correct and the site answered 200), but do not expect a memory-state revert to leave a guest running indefinitely. Reverting the disk and rebooting is the quieter option when the guest does not need to stay up.

Live resizing

`maxvcpus` and `maxmemory` were fixed when the domain was created in M2. That is the whole reason live resize is possible at all.

Baseline before the change:

```
CPU(s):      1
Max memory:   4194304 KiB
Used memory:  2097152 KiB
guest: nproc=1, Mem: 1867 MB
```

```
virsh setvcpus app01 2 --live
virsh setmem app01 3G --live
```

Memory worked immediately and visibly inside the guest, with no reboot:

```
CPU(s):          2
Used memory:     3145728 KiB
guest: nproc=1, Mem: 2891 MB    <-- 1867 -> 2891 MB
```

CPU did **not** work end to end. libvirt reported 2 vCPUs, but the guest still said 1:

```
$ ssh ops@192.168.122.11 'cat /sys/devices/system/cpu/online'    # 0
$ ssh ops@192.168.122.11 'cat /sys/devices/system/cpu/present' # 0-1
$ ssh ops@192.168.122.11 'sudo dmesg | grep -i hot'
[ 130.977897] ACPI: CPU1 has been hot-added
```

So the kernel received the ACPI hotplug event, created the CPU, and left it **offline**. `nproc` counts online CPUs, which is why the guide's verification step as written would have failed on this machine.

The usual reason is that `systemd-udev` online's hot-added CPUs via `40-vm-hotadd.rules`. That file is present and `systemd-udev` is active here, so the uevent was simply missed - a known rough edge of nested virtualization. The fix is one line:

```
echo 1 | sudo tee /sys/devices/system/cpu/cpu1/online # now nproc=2
```

`setvcpus` back down to 1 does work live, so a resize can be undone.

The ceiling is real

This is the interview answer for "how does live resize work". The maximums are fixed at creation and everything happens underneath them:

```
$ virsh setvcpus app01 5 --live
error: invalid argument: requested vcpus is greater than max allowable vcpus
      for the live domain: 5 > 4

$ virsh setmem app01 8G --live
error: invalid argument: cannot set memory higher than max memory
```

Mapping to VMware and Hyper-V

KVM/libvirt is what this lab used. The concepts transfer:

KVM/libvirt (what was done here)	vSphere	Hyper-V
<code>virsh snapshot-create-as</code>	VM snapshot	Checkpoint
<code>virsh snapshot-revert</code>	Revert to snapshot	Apply checkpoint
<code>virsh setvcpus/setmem --live</code>	Hot-add CPU/memory	Dynamic Memory, processor change
qcow2 backing-file overlay	Linked clone	Differencing disk
<code>maxvcpus</code> / <code>maxmemory</code> at creation	VM limits fixed at creation	Startup/maximum RAM
<code>libvirt default</code> NAT network	Standard vSwitch / port group	Virtual switch (NAT/internal)
<code>virsh migrate</code>	vMotion	Live Migration

Being clear that the tooling was KVM and that these are the equivalents is stronger than implying hands-on vSphere experience.

Environment hazard: WSL power-cycles the guests

Worth recording, because it nearly made this module's results meaningless.

WSL shuts down its utility VM when idle. That hard-kills every qemu process, so the guests are power-cycled and their autostart setting brings them back. The guest boot list showed three boots in twelve minutes, and an SSH session died mid-command. A power cut is not a normal way to run a VM, and unflushed guest writes are at risk.

Disabled in `C:\Users\ziaur\.wslconfig`:

```
[wsl2]
vmIdleTimeout=-1
```

Verified rather than assumed: WSL uptime read `25s`, the shell then idled for 150 seconds with no WSL processes running, and the next read was `184s` with both domains still up. Before the change the same idle window restarted the distro.

Two lessons this module produced the hard way:

1. **Absence of evidence is not evidence of absence.** A dropped SSH connection whose stderr was sent to `/dev/null` printed nothing, which looked exactly like “the directory is empty”. That empty-looking listing sent me chasing a lost-write theory for several turns before the real explanation turned up: the delete had worked, and every later playbook run had failed against VMs that were still booting, so nothing was ever redeployed. Checksums at each step are what settled it.
2. **A guard that can never be false is not a guard.** See the `pgrep` truncation bug in [host-setup.md](#).

Backup with a verified restore

Module M6. The claim this module makes is not “there is a backup”. It is “the backup has been restored and the restore matches, and I can prove the checker rejects a bad one”.

The pattern

```
archive the source
-> extract the archive into a scratch directory
-> hash every file in the extracted copy
-> compare against the manifest taken from the source
-> mismatch: print RESTORE MISMATCH and exit non-zero
```

`roles/backup/files/tomcat-backup.sh` does exactly this, and `scripts/backup-failure-drill.sh` proves the failure path works.

Why compare a restore instead of just checksumming the archive

Checksumming the archive only proves the file did not change while it sat there. It says nothing about whether the archive actually contains the deployment it claims to contain, and nothing about whether it is complete.

Comparing a restored tree against the source manifest answers the real question: if I unpacked this on a bare machine, would I get the same bytes back? That is the only property that matters at 3am.

Result on a real run

```
ssh ops@192.168.122.11 'sudo systemctl start tomcat-backup.service'
ssh ops@192.168.122.11 'sudo journalctl -u tomcat-backup.service -n 12 --no-pager'
```

```
RESTORE VERIFIED: 5 files match (/var/backups/tomcat/
webapps-2026-09-30_124302.tar.gz)
Backup complete: /var/backups/tomcat/webapps-2026-09-30_124302.tar.gz (4.0K)
```

```
$ sudo ls -la /var/backups/tomcat/
-rw-r--r-- 1 root root 1327 webapps-2026-09-30_124302.tar.gz
-rw-r--r-- 1 root root 436 webapps-2026-09-30_124302.tar.gz.manifest
-rw-r--r-- 1 root root 99 webapps-2026-09-30_124302.tar.gz.sha256
```

Three files per backup. The archive, the manifest it was verified against, and a checksum of the archive used later for the off-VM copy. Keeping the manifest next to the archive is what makes an independent re-verification possible weeks later, on another machine.

The failure drill

A verifier that only ever prints success proves nothing. The drill tampers with the manifest, expects a non-zero exit, then restores the manifest and expects a pass - so the tool is shown to discriminate rather than to always fail:

```
bash scripts/backup-failure-drill.sh
```

```
Target archive: webapps-2026-09-30_124302.tar.gz

--- manifest tampered, verifying ---
RESTORE MISMATCH: /var/backups/tomcat/webapps-2026-09-30_124302.tar.gz
exit code: 1

--- manifest restored, verifying again ---
RESTORE VERIFIED: 5 files match (/var/backups/tomcat/
webapps-2026-09-30_124302.tar.gz)

DRILL PASSED: the tampered backup was rejected with a non-zero exit code
```

The drill script asserts the outcome, so it fails loudly if the verification ever silently breaks.

The check that exists because of a real incident

The script refuses to run if the source directory has no files:

```
if [ -z "$(manifest "$SRC")" ]; then
    echo "ERROR: $SRC contains no files, refusing to write a meaningless backup" >&2
    exit 1
fi
```

Without it, an empty `webapps/` produces an empty archive and an empty manifest, the diff of two empty manifests passes, and the tool cheerfully prints `RESTORE VERIFIED: 0 files match`. That is a backup that protects nothing while looking healthy.

This is not a hypothetical. During M5 the deployment directory really was empty for a while, and a tool without this guard would have “verified” seven backups of nothing overnight. The guard is cheap; the false confidence is not.

Off-VM copy

```
bash scripts/pull-backups.sh
```

```
receiving incremental file list
webapps-2026-09-30_124302.tar.gz
webapps-2026-09-30_124302.tar.gz.manifest
webapps-2026-09-30_124302.tar.gz.sha256

Verifying the archives survived the transfer:
webapps-2026-09-30_124302.tar.gz: OK
```

The archives land in `~/kvm-lab-backups` on the KVM host, outside the VM they protect, and the checksum computed on the VM is re-checked here. A transfer that corrupted the file could not pass this.

This is still the same physical machine, which is a real limitation: it survives losing the VM, not losing the host. Off-host storage (object storage, or another machine) is the next step for anything that matters, and the README says so plainly rather than implying full disaster recovery.

Scheduling

```
[Timer]
OnCalendar=*-*-* 02:00:00
Persistent=true
```

`Persistent=true` matters here specifically because this lab's VM is frequently switched off. Without it, a missed 02:00 run is simply skipped. With it, `systemd` runs the backup on next boot and catches up.

```
$ systemctl list-timers tomcat-backup.timer
NEXT          LEFT          UNIT          ACTIVATES
Thu 2026-10-01 02:00:00 UTC 13h          tomcat-backup.timer tomcat-backup.service
```

Retention is 7 days, and the `find ... -delete` also removes the orphaned manifests and checksums. An orphaned manifest is worse than no manifest.

Caveat: the guest clock is not trustworthy here

`tar` printed this on every run:

```
tar: ./labapp/work.jsp: time stamp 2026-09-30 17:06:02 is 15767.5 s in the future
```

4.4 hours, which is the offset between this host's timezone and the guest's UTC. The guest's clock jumps around because the VMs get power-cycled and because the M5 memory-state revert rewinds the guest clock (the guest logs `Clock change detected` afterwards). Files written before a jump keep the old timestamp.

Harmless for this backup, because verification compares file contents against a manifest and never looks at timestamps. Worth knowing anyway: do not build anything on guest file mtimes in this environment. If timestamps mattered, the fix is a working guest clock, not a smarter archive.

Monitoring and capacity planning

Module M7. This is the module that produces numbers worth talking about in an interview, so the interesting part is not the table. It is what the table rules out.

Monitoring stack

`docker compose up -d` in `monitoring/` starts Prometheus 3.15.0 and Grafana 13.2.3, both pinned and both with `network_mode: host`.

Host networking is required rather than cosmetic: Prometheus has to reach the VMs on `192.168.122.0/24`, which is libvirt's NAT network inside this distro's network namespace. Verified rather than assumed:

```
$ docker run --rm --network host curlimages/curl:latest \
  -s -o /dev/null -w "%{http_code}\n" http://192.168.122.11:9100/metrics
200
```

Both targets come up:

```
192.168.122.11:9100    job=node    health=up
192.168.122.12:9100    job=node    health=up
```

Four alert rules load and evaluate: `InstanceDown`, `HighCPU`, `LowDisk`, `LowMemory`.

The Grafana datasource is **provisioned from the repo** (`monitoring/grafana/provisioning/`), not clicked into the UI, and the Node Exporter Full dashboard (grafana.com ID 1860) is imported by `monitoring/import-dashboard.sh` through the Grafana API. Nothing in this project depends on someone remembering to click something.

```
$ bash monitoring/import-dashboard.sh
Downloading dashboard 1860 from grafana.com...
468600 bytes
Importing into http://localhost:3000...
{"uid":"rYdddlPWk","title":"Node Exporter Full","imported":true}
```

Tuning the workload first

The original `work.jsp` loop of 3,000,000 iterations cost **196 ms** per request on a 1 vCPU app01 - slower than the 50-100 ms the load test needs, which would have made the server collapse at low concurrency instead of saturating a CPU. Measured, then changed:

```
3,000,000 iterations -> 196.0 ms/request
1,000,000 iterations -> 86.8 ms/request
```

The experiment

`scripts/capacity-test.sh` hot-adds vCPUs, online them inside the guest, runs `ab -n 2000 -c 20` against `/labapp/work` through Apache, and reads peak CPU back from Prometheus. Hot-add works because `create-vms.sh` set `maxvcpus 4` at creation, and the guest has to be told to online the new CPUs (M5).

app01 vCPUs	guest nproc	req/sec	p95	app01 CPU	web01 CPU	host CPU
1	1	17.93	1497 ms	100%	14%	16%
2	2	29.18	971 ms	100%	14%	31%
4	4	44.14	761 ms	100%	14%	56%

Throughput: 1 → 2 vCPUs is a 1.63x gain, 2 → 4 is a further 1.51x. So 4x the CPUs bought 2.46x the throughput, and p95 latency improved 2.0x.

What the table actually rules out

This is the part worth saying out loud, because it is what separates a measured result from a guess:

- **app01 is the bottleneck, and it is confirmed at every size.** Its CPU is pinned at 100% in all three runs. The application really is CPU-bound.
- **The reverse proxy is not the constraint.** web01 sits at 14% regardless of load. A 1 vCPU Apache was never close to being the limit, so the 44 req/sec is not an artefact of the proxy tier.
- **The host is not exhausted.** Host CPU rises 16% → 31% → 56% and never approaches saturation, so the flattening is not caused by running out of host CPU, and `ab` is not the limit either.
- **The host is 4 physical cores with SMT.** An i5-1155G7 reports 4 cores and 8 logical processors, and WSL sees 8 CPUs. At 4 vCPUs, app01 occupies every physical core - sharing them with the load generator, Apache on web01, Prometheus, Grafana and the host itself. Sibling hyperthreads share execution units and L3, which is a good explanation for the 3rd and 4th vCPU adding less than the 1st and 2nd.

What the data does not prove. It cannot separate shared-core contention from the JVM's own serial parts (GC, single-threaded components). Both would produce the same sublinear curve. Telling them apart needs the JMX exporter from the stretch goals, which is exactly what it is for.

Honesty about the numbers

The same experiment was run twice. The first pass, before the CPU readback was fixed, gave 14.71 / 23.88 / 36.65 req/sec; the second gave 17.93 / 29.18 / 44.14. That is a consistent ~20% offset across all three data points, which points at a systematic difference between the runs (JIT warmth, background load) rather than random noise.

So: the **shape** of the result - app01 pinned at 100%, throughput scaling sublinearly, host never saturated - reproduced in both runs and is the finding. The absolute requests per second are indicative, not precise, and the README says so. Quoting 44.14 req/sec as though it were exact would be dishonest.

Background load, checked rather than assumed

`docker ps` revealed three containers from another compose project on this host (`server-operations-lab-api-1`, `-nginx-1`, `-db-1`), running throughout the experiment. Measured rather than hand-waved:

NAME	CPU %	MEM USAGE
server-operations-lab-api-1	0.21%	39.09MiB
server-operations-lab-nginx-1	0.00%	6.84MiB
server-operations-lab-db-1	0.00%	19.28MiB

Effectively idle, so they did not distort these results. Worth recording anyway, because “the numbers looked fine so the environment must have been clean” is how capacity tests go wrong.

What I would do next

- Move the load generator off the host. `ab` shares 4 physical cores with the thing it is measuring, which caps how much the vCPU count can help.
- Give web01 more than 1 vCPU so the proxy tier is excluded at higher load rather than merely observed to be low at this load.
- Add the JMX exporter to separate JVM GC behaviour from raw CPU saturation.
- Run a longer test with a mixed workload (`/health` and `/work`) to see whether the ceiling is per-request CPU or connection handling.

Limitations

Single host, single app VM, no HA. The absolute numbers describe a laptop CPU running nested virtualisation, not a data centre. They are useful for demonstrating the method and the shape of the scaling curve, not for sizing real hardware.

Firewall hardening

Module M8, optional in the project guide, and worth doing because it turns the two-tier design from a diagram into an enforced fact.

What the rules encode

The design says: the world reaches only the web tier, the web tier reaches the application, and only the monitoring host may scrape. The rules say exactly that.

app01 (application tier)

Port	Allowed from	Why
22/tcp	anywhere	administration
9100/tcp	192.168.122.1	node exporter, monitoring host only
8080/tcp	192.168.122.12	Tomcat, web tier only

web01 (web tier)

Port	Allowed from	Why
22/tcp	anywhere	administration
9100/tcp	192.168.122.1	node exporter, monitoring host only
80/tcp	anywhere	the only ingress to the application

Both: `default deny (incoming)`.

Ordering is the safety property

The role applies every allow rule **before** switching on the default-deny policy:

```
- name: Allow SSH
  community.general.ufw: {rule: allow, port: "22", proto: tcp}
# ... all other allow rules ...
- name: Default deny incoming and enable
  community.general.ufw: {state: enabled, policy: deny, direction: incoming}
```

If those were reversed, the first thing an empty default-deny ruleset would kill is the SSH session running the playbook. `ansible-playbook` does not survive having its own connection dropped, and a lab VM that cannot be reached is a manual console recovery. This is the cheapest possible insurance.

A snapshot (`clean-deploy`) and `virsh console` were both available as recovery paths before the role was run, which is why it was safe to run at all.

Verification

Each claim checked rather than asserted:

The proxy path still works.

```
through Apache: HTTP 200 body=OK app01
```

Direct access to Tomcat from the host is now blocked. This is the change that matters, because before hardening the same request returned `HTTP 200`:

```
app01:8080 from host -> HTTP 000 (exit 28)
```

`curl` exit code 28 is a timeout, not a refusal - the firewall is dropping the packets rather than rejecting them, which is the correct behaviour for a default-deny policy. Confirmed from the other side too: `dmesg` on `app01` showed **10 UFW log lines** for the attempt. A blocked connection that leaves no log is indistinguishable from a broken service.

Monitoring was not broken by the firewall. This is the classic self-inflicted wound with `ufw`, and it is why the 9100 rule names the monitoring host explicitly:

```
192.168.122.11:9100    up
192.168.122.12:9100    up
```

SSH is key-only.

```
$ ssh -o PubkeyAuthentication=no ops@192.168.122.11
ops@192.168.122.11: Permission denied (publickey).
```

Password authentication was already disabled in `cloud-init` (`ssh_pwauth: false`), so this confirms the setting rather than being the thing that created it.

Idempotency survived. The new role reports `changed=0` on the next two runs:

```
app01 : ok=18 changed=0 failed=0
web01 : ok=17 changed=0 failed=0
```

That is the full role set now - `common`, `node_exporter`, `tomcat`, `backup`, `apache`, `hardening` - still converging to no changes.

Why this is more than a checkbox

Before this module, `curl http://192.168.122.11:8080/labapp/health` from the host worked, which means the reverse proxy was decorative: anything that could reach the network could bypass Apache and talk to Tomcat directly. After it, the only way in is port 80 on `web01`.

That is the actual justification for a reverse proxy tier - one ingress point, one place for access logs, TLS termination, load balancing, and the app server not directly reachable - and this module is where that stops being a claim in a README and becomes something a packet cannot argue with.

What this still does not do

Being clear about the gaps is the difference between hardening and theatre:

- **SSH is open to anywhere.** A real environment would restrict it to a bastion host or an admin range. Left open here deliberately so the lab stays reachable.
 - **No brute-force protection.** `fail2ban` or key-only plus rate limiting would be the next layer; key-only auth makes the risk much smaller but not zero.
 - **No TLS.** Port 80 is plaintext. The guide's stretch goal covers a self-signed certificate on Apache.
 - **Host-level firewall, not network-level.** Real segmentation belongs in security groups or network ACLs, so a misconfigured host cannot open itself up. `ufw` is the last line, not the only one.
 - **Egress is unrestricted.** Default deny applies to incoming only. Limiting outbound traffic is a separate project.
 - **It does nothing for availability.** Firewall rules are not redundancy; a failed app01 is still a failed site.
-

Disaster recovery

Module M9. This is the drill that justifies everything in [backup-and-restore.md](#): it proves the backup restores onto a machine that never had the application, and it produces a measured recovery time instead of a guess.

Measured result

What	Time
Total VM loss → working service	348 s (5 min 48 s)
Deployment content restored and verified	8 s

“Working service” means `curl http://192.168.122.12/labapp/health` returning `200 OK app01` through the reverse proxy, on a VM rebuilt from nothing.

The drill

Starting point: one verified backup pulled off the VM (`~/kvm-lab-backups`), and `app01` running with the application deployed.

1. Total loss

```
virsh destroy app01
virsh undefine app01 --remove-all-storage --snapshots-metadata
```

```
Domain 'app01' destroyed
Domain 'app01' has been undefined
Volume 'vda'(/var/lib/libvirt/images/app01.qcow2) removed.

$ virsh list --all
 Id   Name    State
-----
 2    web01   running
```

The disk file is gone. So is the `clean-deploy` snapshot, because a snapshot lives on the same disk it protects - which is the whole point of the snapshot-versus-backup distinction, demonstrated rather than described.

2. Rebuild the machine

```
bash scripts/create-vms.sh
```

`create-vms.sh` is idempotent, so it skipped `web01` and rebuilt only `app01` from the shared cloud image - a genuinely fresh disk, not a repaired one. The guest was reachable over SSH about 20 seconds later.

3. Configure it

```
ansible-playbook site.yml
```

```
PLAY RECAP
app01  : ok=18   changed=11   unreachable=0   failed=0
web01  : ok=17   changed=0    unreachable=0   failed=0
```

changed=11 on app01 versus changed=0 on web01 is the useful signal: the playbook did real work on the blank machine and nothing at all on the untouched one. Time to a working service: **348 s**.

4. Restore the deployment content

The rebuilt machine had Tomcat installed by the playbook, but the off-VM archive is the independent copy, so it gets exercised too. Deployment destroyed on the fresh VM:

```
$ ssh ops@192.168.122.11 'sudo rm -rf /var/lib/tomcat10/webapps/*'
through Apache now: HTTP 404
```

Then restored from the host-side copy, verified with the same tool that made it:

```
$ scp ~/kvm-lab-backups/webapps-2026-09-30_132943.tar.gz* ops@192.168.122.11:/tmp/
$ ssh ops@192.168.122.11 'sudo /usr/local/bin/tomcat-backup.sh verify /tmp/
webapps-2026-09-30_132943.tar.gz'
RESTORE VERIFIED: 5 files match (/tmp/webapps-2026-09-30_132943.tar.gz)

$ ssh ops@192.168.122.11 'sudo tar -xzf /tmp/webapps-*.tar.gz -C /var/lib/tomcat10/
webapps
                                && sudo chown -R tomcat:tomcat /var/lib/tomcat10/webapps
                                && sudo systemctl restart tomcat10'

content restored: 8s

health after restore: HTTP 200  body=OK app01
WAR on the restored VM: 03ab3c83afc5fad7...
WAR in the repo:       03ab3c83afc5fad7...
```

The WAR checksum on the rebuilt machine matches the repo exactly, so the bytes came back intact rather than merely coming back.

What the drill taught me

Two recovery paths, and they cover different things. The **machine** comes from the cloud image plus Ansible. The **application content** comes from the backup. Neither alone is a recovery plan, and the 348 s figure is dominated by the first path.

The backup scope is the deployment, not the machine. The backing-up tree was `/var/lib/tomcat10/webapps`, so Tomcat's own default `ROOT` application is not in it. After the restore, `/labapp/*` works and `/` does not. That is correct for this design - the proxy only needs the application - but it is a scope decision worth stating out loud rather than discovering during an incident.

The backup contains a Tomcat artifact. The five archived files are `labapp.war`, `labapp/health.jsp`, `labapp/work.jsp`, `labapp/WEB-INF/web.xml` and `labapp/META-INF/war-tracker`. That last one is Tomcat's own deployment tracker, generated by Tomcat, not written by me. It is harmless here and it made the verification identical on both machines, but a stricter backup would exclude generated metadata so that a restore cannot confuse a redeployment.

In this lab the backup is not the only copy. The WAR is also in git, so the playbook alone restored a working service in 348 s. The backup proves the independent path works, but for a truer test of "the backup was all I had", the application would need state that git does not hold - a database, uploaded files, unmanaged config. Saying this is more useful than implying the backup saved the day.

Reproducing

Steps 2 and 3 are safe and repeatable at any time. Step 1 is the only irreversible action in this project; it is safe here only because the backup is already off the VM and verified, and because `create-vm.sh` rebuilds the machine from a cloud image.

Interview notes

The questions this project prepares you for, with the answer grounded in something this lab actually measured. Replace nothing with theory; every claim below has output behind it in [evidence/](#).

1. Walk me through your backup verification. Archive the deployment, extract it into a scratch directory, hash every file in the restore, and compare that against a manifest taken from the source. Any difference prints `RESTORE MISMATCH` and exits non-zero, so a scheduled backup that fails is a failed job rather than a green tick. I compare the restore rather than the archive because a checksum of the archive only proves the file did not change - it says nothing about whether the archive contains the deployment. Evidence: `evidence/04-backup-failure-drill.txt`.

2. Snapshot versus backup? A snapshot is point-in-time state on the same storage as the VM, for fast rollback before a change. A backup is an independent, verified copy elsewhere, for disaster recovery. A snapshot does not survive losing the disk - and I can show that: the recovery drill destroys the VM with `--remove-all-storage` and the snapshot dies with it. `docs/disaster-recovery.md`.

3. What does idempotent mean, and how did you prove it? Running the playbook twice changes nothing the second time. The proof is the second `PLAY RECAP`: `changed=0` on both hosts, with `ok=18` tasks on `app01` - so the tasks ran and reported nothing to do, rather than being skipped. I also found that claim was quietly false for a while, because rebuilding an unchanged WAR produced a different file and the deploy task reported `changed` every run; `zip` extra metadata was the cause. `evidence/01-idempotency.txt`.

4. A user says the site is down. What do you check? Apache up, then whether the proxy can reach Tomcat, then whether Tomcat is healthy, then logs (Apache vhost logs, `journalctl -u tomcat10`), then firewall, then disk and memory. The useful distinction: a `404` from the proxy means Tomcat is fine and the application is missing, while a `503` means the app server itself is unreachable. I hit both during this build. `evidence/02`.

5. How did you find the bottleneck? Load-tested through the proxy while reading CPU back from Prometheus. `app01` was pinned at 100% at 1, 2 and 4 vCPUs, so the application was CPU-bound. `web01` sat at 14% and the host never passed 56%, which ruled out the proxy tier and host saturation as explanations for the sublinear scaling - 4x the CPUs bought 2.46x the throughput. `docs/capacity-planning.md`.

6. How does live resize work? Maximums are fixed when the VM is created, and everything happens below them. `setvcpus` past `maxvcpus` fails with `5 > 4`, and `setmem` past `maxmemory` fails with `cannot set memory higher than max memory`. That is why `create-vm.sh` sets `maxvcpus 4` at creation. Memory hot-added and was visible in the guest immediately. The vCPU hot-add needed an extra step: `libvirt` reported 2 vCPUs, the kernel logged `ACPI: CPU1 has been hot-added`, and the CPU was left offline until I online'd it, so `nproc` still said 1. `evidence/07-live-resize.txt`.

7. How would this differ on vSphere or Hyper-V? Use the mapping table in the README - snapshot/checkpoint, hot-add, linked clone/differencing disk, the NAT network

versus vSwitch or port group, vMotion versus `virsh migrate`. Be clear that the tooling used here was KVM.

8. Why a reverse proxy in front of Tomcat? TLS termination, load balancing, static content, one place for access logs, and hiding the app server. This lab makes the last one concrete: before hardening, `curl` straight to `app01:8080` from the host returned HTTP 200, which meant the proxy was decorative and anything on the network could bypass it. After the `ufw` rules, the only way in is port 80 on `web01`. `docs/hardening.md`.

9. What would you add for real high availability? A second app VM with Apache `mod_proxy_balancer` or HAProxy health checks, a shared or replicated database, and a floating IP such as `keepalived` on the proxy tier. Also worth naming: this lab has no HA at all, and a firewall is not redundancy.

10. How would you handle WebLogic or WebSphere? I have not used either. The concepts carry - domains and clusters, deployment, JVM tuning, thread pools - and the admin tooling is what would need learning.

The two questions that actually decide it

“Tell me about something that broke.” Have three ready, because they are the strongest material in the project:

1. A guard I wrote used `pgrep -x qemu-system-x86_64`, which can never match because Linux truncates process names to 15 characters. The negation was always true, so the “safety” check would have deleted the network bridge out from under running VMs. It looked correct. Only testing it against a live VM caught it.
2. My idempotency proof was false until I noticed that `zip` writes extra metadata, so rebuilding an unchanged artifact produced a different file.
3. Guests were being power-cycled between commands and my work kept vanishing. I ruled out the idle timeout, `systemd-oomd`, the watchdog and a daemon crash before noticing that kernel boot messages repeated while `boot_id` and uptime stayed constant - which meant the WSL kernel was alive and only the distro was being torn down.

“What are the limitations?” The README lists seven. Being able to name them is worth more than pretending the lab is production-grade.

Build log

One entry per module. Raw commands, real output, and what broke.

Module	What it delivers	Status
M0	Repository scaffold	done
M1	KVM host install + validation	done
M2	Two VMs with fixed IPs	done
M3	Sample app WAR	done
M4	Ansible roles + idempotency proof	done
M5	Snapshots + live resize	done
M6	Verified backup + restore drill	done
M7	Monitoring + capacity plan	done
M8	Firewall hardening (optional)	done
M9	README, screenshots, resume	done

Environment

- Host: Windows 11 Pro, 20 GB RAM, 8 cores exposed to WSL2
- Guest host: WSL2 Ubuntu 24.04.3 LTS, systemd PID 1, `/dev/kvm` present
- Repo lives at `~/kvm-tomcat-lab` inside WSL (native ext4, not `/mnt/c`)
- Git remote: `git@github.com:ziaur390/kvm_tomacat.git`

M0 - Repository scaffold

`git init -b main`, committed `.gitignore`, README stub and `docs/screenshots/`. Direct push to `main` for the initial commit only; every module from M1 onward lands through a feature branch and a pull request.

M1 - KVM host install + validation

Detailed notes: [host-setup.md](#)

- Installed `qemu`, `libvirt`, `virtinst`, `cpu-checker`, `zip`; enabled `libvirtd`.
- `kvm-ok` reports “KVM acceleration can be used” - nested virt works in WSL2.
- Added the user to `libvirt` and `kvm`; restarted the distro to apply groups. `virsh` then works without `sudo`.
- Downloaded and checksum-verified the noble cloud image (597 MB).

What broke

After the distro restart the `default` network would not start: `Network is already in use by interface virbr0`. WSL keeps the kernel alive across a distro restart, so `virbr0` survived while `libvirtd` did not, and the new daemon will not adopt a bridge it did not create.

Fix: `sudo ip link delete virbr0 && virsh net-start default`. Documented in [host-setup.md](#) because it will recur after every WSL restart.

M2 - Two VMs with fixed IPs

- Generated `~/.ssh/kvmlab` (ed25519, no passphrase - lab only).
- `scripts/create-vm.sh` reserves the two addresses by MAC in libvirt's DHCP and creates both domains from the cloud image, with `maxvcpus/maxmemory` set for the later live resize in M5.
- Both VMs boot, get their reserved addresses, and accept key-only SSH as `ops`.
- `ansible all -m ping` succeeds against both.

```
$ virsh list --all
Id   Name    State
1    app01   running
2    web01   running

$ virsh net-dhcp-leases default
192.168.122.11/24    52:54:00:aa:00:11    app01
192.168.122.12/24    52:54:00:aa:00:12    web01

$ ssh ops@192.168.122.11 'hostname && nproc && free -m | head -2'
app01
1
Mem:  1867 total  1433 free
```

The 2048 MiB is less than the recommended 3072 MiB warning from `virt-install` is deliberate: `app01` is undersized so the M7 load test has a bottleneck.

What broke

1. The stale `virbr0` bit again while installing packages, so the reservations step failed with `network is not running`. Because it kept recurring, the one-line fix became a permanent one: a systemd `ExecStartPre` running `scripts/libvirt-clear-stale-bridge.sh`, guarded so it only fires when no qemu domain is running.
2. `ansible-galaxy collection install community.general` pulled 13.4.0 into `~/.ansible/collections`, which warns that it does not support `ansible-core 2.16.3` - and that user path shadows the supported 8.3.0 that the apt `ansible` package already ships in `/usr/lib/python3/dist-packages/ansible_collections`. Removed the user copy; no galaxy install is needed on this host.

M3 - Sample app WAR

Three tiny files and a build script:

- `app/src/health.jsp` - returns `OK <hostname>`, used by Ansible's deployment health check and by every "is it up" question later.
- `app/src/work.jsp` - burns CPU in a loop on purpose. M7's load test needs an endpoint that actually saturates a vCPU; a page that just prints text would leave nothing to measure.
- `app/src/WEB-INF/web.xml` - maps `/health` and `/work` onto those JSPs, so the URLs do not end in `.jsp`.
- `app/build.sh` - zips `src/` into `roles/tomcat/files/labapp.war`.

```
$ bash app/build.sh
built ../../roles/tomcat/files/labapp.war
  Length      Name
     96      health.jsp
    162      work.jsp
    554    WEB-INF/web.xml
```

`build.sh` fails if `WEB-INF/web.xml` is missing from the archive. Without that guard the symptom in Tomcat is a confusing 404 rather than a broken build.

Deploys as context `/labapp`, so the endpoints are `/labapp/health` and `/labapp/work`. A named context avoids fighting Tomcat's built-in ROOT app.

TODO (M7): the loop count of 3,000,000 is a guess. Measure the response time under `ab` and tune it so one request costs roughly 50-100 ms.

M4 - Ansible roles, deploy and idempotency proof

Four roles: `common` (apt cache, curl, http), `node_exporter` (the metrics agent for M7), `tomcat` (WebLogic-style app tier), `apache` (reverse proxy).

First run

```
PLAY RECAP
app01  : ok=10  changed=4  failed=0
web01  : ok=14  changed=9  failed=0
```

The Tomcat health check retried twice before passing. That is expected, not a bug: the first request to a JSP makes Tomcat compile it, which takes a few seconds. Hence `retries: 12` `delay: 5` rather than a single attempt.

The deployment works

```
$ curl http://192.168.122.12/labapp/health    # through Apache on web01
OK app01

$ curl http://192.168.122.11:8080/labapp/health # direct to Tomcat
OK app01
```

Two things worth noting. The response is `OK app01` while the URL pointed at `web01` - the hostname in the body is the app machine's, which proves the request really was proxied rather than served locally. And the same URL works directly on 8080, which is what M8 will close off with `ufw`.

Second run - idempotency proof

```
PLAY RECAP
app01  : ok=10   changed=0   failed=0
web01  : ok=12   changed=0   failed=0
```

`changed=0` means every task checked the actual state of the machine and found nothing to do. That is what idempotent means, and it is the difference between configuration management and a shell script.

What broke

1. **A rebuilt WAR was never byte-identical.** Running `app/build.sh` twice with no source changes produced two different archives, so the `Deploy the lab WAR` copy task reported `changed` on every run and the idempotency proof was a lie waiting to be exposed. Cause: Info-ZIP writes extra metadata (including high-precision timestamps) into the archive. Fix: `zip -X`.

```
before -X: d6432034986adb0e...
after  -X: 2e2a19812c6e926f...
```

After the fix, two rebuilds hash identically, and the playbook reports `changed=1` for the new WAR and then `changed=0` on the run after that. Worth knowing: an idempotency claim is only as good as the artifact you feed it.

2. `a2disssite` replaced the guide's "delete the symlink with the `file` module" step. Deleting a symlink by path is a footgun, and the `removes:` guard makes the real tool report `ok` instead of `changed` on later runs.

M5 - Snapshots, rollback and live resizing

Full write-up: [snapshots-and-resize.md](#)

- Took a `clean-deploy` snapshot, recorded the WAR sha256, deleted the deployed app (HTTP 404 through Apache), reverted, and confirmed the WAR returned with the identical hash.
- The revert took 5.1 s and did **not** reboot the guest (uptime `up 1 minute` before and after), so it is a memory-state rollback rather than a disk restore plus boot.
- Live resize: memory `1867 MB -> 2891 MB` inside the guest with no reboot. Two vCPUs were added at the hypervisor but the guest kept `nproc=1` - the kernel logged `ACPI: CPU1 has been hot-added and left it offline`, so it had to be onlined by hand. The guide's `nproc` verification step would have failed here as written.
- Exceeding the creation-time ceiling fails loudly: `setvcpus 5` and `setmem 8G` are both rejected with errors naming the maximums.

What broke

1. **The WSL idle shutdown was power-cycling the guests**, hard-killing qemu and restarting the VMs. Three boots in twelve minutes, plus an SSH session that died mid-command. Fixed with `vmIdleTimeout=-1` in `C:\Users\ziaur\.wslconfig` and verified with a 150 second idle test (uptime `25s -> 184s`).
2. **The stale-bridge guard could never be false.** `pgrep -x qemu-system-x86_64` never matches because Linux truncates process names to 15 characters, so the negation was always true and the script would have deleted `virbr0` on every `libvirtd` start, including with live VMs. Replaced with a check of the bridge's own port list.
3. **A dropped SSH connection silently looked like an empty directory.** The destructive `rm` had worked, but the listing before it printed nothing because the connection dropped and stderr went to `/dev/null`. That sent me chasing a lost-write theory for several turns. The real story was duller: every later playbook run failed against guests that were still booting, so nothing was ever redeployed. Checksums at every step settled it. Lesson: do not swallow stderr on the command that is supposed to prove something.

M6 - Backup with a verified restore

Full write-up: [backup-and-restore.md](#)

- `tomcat-backup.sh` archives `webapps/`, extracts the archive into a scratch directory, hashes the restore, compares it against a manifest of the source, and exits non-zero on any disagreement.
- Real run: `RESTORE VERIFIED: 5 files match`, plus a 7-day retention sweep and a nightly systemd timer with `Persistent=true`.
- `scripts/backup-failure-drill.sh` tampers with a manifest, asserts the verifier rejects it with exit code 1, then restores the manifest and asserts it passes. Both outcomes are asserted, so the drill fails if the verification ever silently breaks.
- `scripts/pull-backups.sh` rsyncs the archives to the KVM host and re-checks the checksum that was computed on the VM.

Deliberate addition beyond the guide

The script refuses to back up an empty source directory. Two empty manifests compare equal, so the guide's version as written would print `RESTORE VERIFIED: 0 files match` for a deployment directory containing nothing. That is not theoretical: `webapps/` really was empty for a while during M5, and an unguarded tool would have "verified" a night of meaningless backups.

Environment note: a persistent WSL session is required

Root cause of the churn in M5, finally pinned down. WSL tears down the distro when the last session exits. That stops `libvirtd`, which takes qemu with it, so the guests are power-cycled and only come back because of autostart.

Evidence: the kernel messages `Linux version 6.18.40.1-microsoft-standard-WSL2` and `mini_init: WSL user cgroup created` repeat between commands, while `/proc/uptime` and

`boot_id` carry on regardless - because the WSL VM kernel stays alive and only the distro is torn down. `vmIdleTimeout=-1` does not help, because this is session-driven, not idle-driven. Workaround, verified: hold a session open.

```
nohup wsl.exe -e bash -c "sleep 3600" >/dev/null 2>&1 &
```

With that running, qemu process start times and `libvirtd`'s start time stayed identical across a 90 second gap and the guests stopped rebooting. Without it, the guests were rebooted between almost every command.

Follow-up after the revert

Roughly a minute after the memory-state revert the guest rebooted by itself, the last log line being `systemd-resolved: Clock change detected`. A memory restore rewinds the guest clock and invalidates in-flight I/O, so the guest notices.

Ruled out the scary explanation: the guest does **not** arm the itco watchdog (`RuntimeWatchdogUsec=0`), so a watchdog reset is not the cause. That mattered to check, because an armed watchdog firing under load would have silently corrupted the M7 capacity numbers.

M7 - Monitoring and capacity planning

Full write-up: [capacity-planning.md](#)

- Prometheus 3.15.0 and Grafana 13.2.3 pinned in `monitoring/docker-compose.yml` with host networking, both targets `health=up`, four alert rules evaluating.
- The Grafana datasource is provisioned from the repo and dashboard 1860 is imported through the API by `monitoring/import-dashboard.sh`, so no step needs a human to click something.
- Tuned `work.jsp` from 3,000,000 loop iterations (196 ms/request) to 1,000,000 (86.8 ms/request), inside the 50-100 ms the load test needs.
- `scripts/capacity-test.sh` sweeps 1, 2 and 4 vCPU and reads peak CPU back from Prometheus.

app01 vCPUs	req/sec	p95	app01 CPU	web01 CPU	host CPU
1	17.93	1497 ms	100%	14%	16%
2	29.18	971 ms	100%	14%	31%
4	44.14	761 ms	100%	14%	56%

4x the CPUs bought 2.46x the throughput. app01 is pinned at 100% at every size, so the application is genuinely CPU-bound; web01 at 14% and the host below 60% rule out the proxy tier and host saturation as explanations for the flattening. The host turns out to be 4 physical cores with SMT, and app01's 4 vCPUs occupy all of them alongside the load generator.

What the data cannot separate: shared-core contention from the JVM's own serial parts (GC, single-threaded components). Both produce the same sublinear curve. The JMX exporter from the stretch goals is what would tell them apart.

What broke

1. The first pass of the experiment reported blank CPU columns. The readback used a Python f-string with escaped quotes inside a double-quoted shell string, which Python rejected as a line continuation.
2. The same experiment run twice differed by a consistent ~20% per data point (14.71/23.88/36.65 then 17.93/29.18/44.14), which is systematic rather than noise. The shape reproduced; the absolute figures are indicative only, and the README says so rather than quoting 44.14 req/sec as though it were exact.
3. `docker ps` showed three containers from another compose project on the host (`server-operations-lab-*`), running throughout the tests. Measured instead of assumed: 0.00-0.21% CPU, so no distortion.
4. Grafana looked dead on `localhost:3000` for about a minute after `compose up`, then answered normally. It was still installing bundled plugins. Retry before debugging.

M8 - Firewall hardening

Full write-up: [hardening.md](#)

app01 allows	from	web01 allows	from
22/tcp	anywhere	22/tcp	anywhere
9100/tcp	192.168.122.1	9100/tcp	192.168.122.1
8080/tcp	192.168.122.12	80/tcp	anywhere

Both default-deny incoming. Allow rules are applied before the default-deny policy is switched on, because an empty deny ruleset would kill the SSH session running the playbook.

Before hardening, `curl http://192.168.122.11:8080/labapp/health` from the host returned `HTTP 200`, which made the reverse proxy decorative. After it:

```
through Apache:      HTTP 200  body=OK app01
app01:8080 from host: HTTP 000 (curl exit 28 - dropped, not refused)
dmesg on app01 for the attempt: 10 UFW log lines
```

Prometheus targets stayed `up`, which is the usual self-inflicted casualty of a `ufw` rollout and the reason the 9100 rule names the monitoring host explicitly. Key-only SSH confirms with `Permission denied (publickey)`. The full role set still converges: `changed=0` on the next two runs.

Gaps recorded in the write-up rather than glossed over: SSH open to anywhere, no TLS, no `fail2ban`, host-level rather than network-level enforcement, unrestricted egress, and no effect on availability.

M9 - README, evidence and the disaster recovery drill

- `scripts/capture-evidence.sh` regenerates `docs/evidence/` from a live lab, so every number in the README can be checked or regenerated. Text rather than screenshots wherever the proof is textual: greppable and diffable.
- Full README written from the measured results, plus [disaster-recovery.md](#) and [interview-notes.md](#).

The drill

```
virsh destroy app01
virsh undefine app01 --remove-all-storage --snapshots-metadata # disk gone,
snapshot gone with it
bash scripts/create-vms.sh # fresh machine from
the cloud image
ansible-playbook site.yml # app01 changed=11,
web01 changed=0
```

- **348 s (5 min 48 s)** from total VM loss to `200 OK app01` through the proxy.
- The off-VM archive then verified on the rebuilt machine with the same tool that created it: `RESTORE VERIFIED: 5 files match`, and extraction plus a Tomcat restart took **8 s** to `200 OK app01`.
- The WAR checksum on the rebuilt machine matched the repo exactly.

What the drill taught

1. **Two recovery paths covering different things.** The machine comes from the cloud image plus Ansible; the application content comes from the backup. The 348 s is dominated by the first path.
2. **The backup scope is the deployment, not the machine.** Tomcat's default `R00T` app is not in it, so after a restore `/labapp/*` works and `/` does not. Correct for this design, but a scope decision worth stating.
3. **The archive contains a Tomcat artifact.** One of the five files is `labapp/META-INF/war-tracker`, generated by Tomcat rather than written by me. Harmless here; a stricter backup would exclude generated metadata.
4. **In this lab the backup is not the only copy.** The WAR is also in git, so the playbook alone restored a working service. For a truer test the app would need state git does not hold - a database, uploads, unmanaged config.
5. `clean-deploy` was re-taken on the rebuilt VM so the lab matches its documentation again, and the evidence was re-captured against it.

Appendix: raw evidence

Command output captured from the running lab. Regenerate with `bash scripts/capture-evidence.sh`.

01-idempotency

```
### Two consecutive ansible-playbook runs. The second must report changed=0.
### captured 2026-09-30T13:39:13Z

$ ansible-playbook site.yml # run 1
PLAY RECAP *****
app01          : ok=18  changed=0    unreachable=0    failed=0
skipped=1      rescued=0  ignored=0
web01          : ok=17  changed=0    unreachable=0    failed=0
skipped=1      rescued=0  ignored=0

$ ansible-playbook site.yml # run 2
PLAY RECAP *****
app01          : ok=18  changed=0    unreachable=0    failed=0
skipped=1      rescued=0  ignored=0
web01          : ok=17  changed=0    unreachable=0    failed=0
skipped=1      rescued=0  ignored=0
```

02-proxy-vs-direct

```
### The reverse proxy is load-bearing: the app answers through web01 and is
unreachable directly from the host, which is not on the allow list.
### captured 2026-09-30T13:39:58Z

$ curl http://192.168.122.12/labapp/health          # through Apache on web01
OK app01

$ curl http://192.168.122.11:8080/labapp/health    # direct to Tomcat, bypassing the
proxy
HTTP 000 (curl exit 28)
  exit 28 = timed out, i.e. dropped by the firewall rather than refused

$ sudo ufw status verbose    # on app01
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                Action      From
--                -
22/tcp            ALLOW IN    Anywhere
9100/tcp          ALLOW IN    192.168.122.1
8080/tcp          ALLOW IN    192.168.122.12
22/tcp (v6)       ALLOW IN    Anywhere (v6)

$ sudo ufw status    # on web01
Status: active

To                Action      From
--                -
22/tcp            ALLOW       Anywhere
9100/tcp          ALLOW       192.168.122.1
80/tcp            ALLOW       Anywhere
22/tcp (v6)       ALLOW       Anywhere (v6)
80/tcp (v6)       ALLOW       Anywhere (v6)
```

03-backup-restore-verified

```
### Backing up shows the restore being verified, not just the archive created.
### captured 2026-09-30T13:40:05Z

Sep 30 13:40:05 app01 systemd[1]: Starting tomcat-backup.service - Tomcat webapps
backup with restore verification...
Sep 30 13:40:05 app01 tomcat-backup.sh[9049]: tar: ./labapp/work.jsp: time stamp
2026-09-30 18:00:08 is 15602.34993012 s in the future
Sep 30 13:40:05 app01 tomcat-backup.sh[9049]: tar: ./labapp/health.jsp: time stamp
2026-09-30 17:06:02 is 12356.349421775 s in the future
Sep 30 13:40:05 app01 tomcat-backup.sh[9049]: tar: ./labapp/WEB-INF/web.xml: time
stamp 2026-09-30 17:06:02 is 12356.349185671 s in the future
Sep 30 13:40:05 app01 tomcat-backup.sh[9026]: RESTORE VERIFIED: 5 files match (/var/
backups/tomcat/webapps-2026-09-30_134005.tar.gz)
Sep 30 13:40:05 app01 tomcat-backup.sh[9026]: Backup complete: /var/backups/tomcat/
webapps-2026-09-30_134005.tar.gz (4.0K)
Sep 30 13:40:05 app01 systemd[1]: tomcat-backup.service: Deactivated successfully.
Sep 30 13:40:05 app01 systemd[1]: Finished tomcat-backup.service - Tomcat webapps
backup with restore verification.

$ sudo ls -la /var/backups/tomcat/
total 20
drwxr-x--- 2 root root 4096 Sep 30 13:40 .
drwxr-xr-x 3 root root 4096 Sep 30 13:40 ..
-rw-r--r-- 1 root root 1637 Sep 30 13:40 webapps-2026-09-30_134005.tar.gz
-rw-r--r-- 1 root root 436 Sep 30 13:40 webapps-2026-09-30_134005.tar.gz.manifest
-rw-r--r-- 1 root root 99 Sep 30 13:40 webapps-2026-09-30_134005.tar.gz.sha256

$ systemctl list-timers tomcat-backup.timer
NEXT                                LEFT LAST PASSED UNIT                                ACTIVATES
Thu 2026-10-01 02:00:00 UTC 12h -                                - tomcat-backup.timer tomcat-
backup.service

1 timers listed.
Pass --all to see loaded but inactive timers, too.
```

04-backup-failure-drill

```
### Proof the verification discriminates: a tampered manifest is rejected with
a non-zero exit code, and the same archive passes again once restored.
### captured 2026-09-30T13:40:09Z

Warning: Permanently added '192.168.122.11' (ED25519) to the list of known hosts.
Target archive: webapps-2026-09-30_134005.tar.gz

--- manifest tampered, verifying ---
tar: ./labapp/work.jsp: time stamp 2026-09-30 18:00:08 is 15598.399522598 s in the
future
tar: ./labapp/health.jsp: time stamp 2026-09-30 17:06:02 is 12352.399291756 s in the
future
tar: ./labapp/WEB-INF/web.xml: time stamp 2026-09-30 17:06:02 is 12352.399182449 s in
the future
RESTORE MISMATCH: /var/backups/tomcat/webapps-2026-09-30_134005.tar.gz
exit code: 1

--- manifest restored, verifying again ---
tar: ./labapp/work.jsp: time stamp 2026-09-30 18:00:08 is 15598.380171162 s in the
future
tar: ./labapp/health.jsp: time stamp 2026-09-30 17:06:02 is 12352.379914855 s in the
future
tar: ./labapp/WEB-INF/web.xml: time stamp 2026-09-30 17:06:02 is 12352.379795802 s in
the future
RESTORE VERIFIED: 5 files match (/var/backups/tomcat/
webapps-2026-09-30_134005.tar.gz)

DRILL PASSED: the tampered backup was rejected with a non-zero exit code
```

05-offvm-copy

```
### The backup exists outside the VM it protects, and the checksum computed on
the VM is re-checked after the transfer.
### captured 2026-09-30T13:40:09Z

@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@   WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!   @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
IT IS POSSIBLE THAT SOMEONE IS DOING SOMETHING NASTY!
Someone could be eavesdropping on you right now (man-in-the-middle attack)!
It is also possible that a host key has just been changed.
The fingerprint for the ED25519 key sent by the remote host is
SHA256:k2j/yqMoXzgybpYch5wvmKWJXUBBFhsWYUfrLpScXm0.
Please contact your system administrator.
Add correct host key in /home/ziaur26261/.ssh/known_hosts to get rid of this message.
Offending ECDSA key in /home/ziaur26261/.ssh/known_hosts:12
  remove with:
    ssh-keygen -f '/home/ziaur26261/.ssh/known_hosts' -R '192.168.122.11'
Host key for 192.168.122.11 has changed and you have requested strict checking.
Host key verification failed.
rsync: connection unexpectedly closed (0 bytes received so far) [Receiver]
rsync error: unexplained error (code 255) at io.c(232) [Receiver=3.2.7]

$ ls -la ~/kvm-lab-backups/
total 44
drwxr-x---  2 ziaur26261 ziaur26261 4096 Sep 30 18:29 .
drwxr-x--x 17 ziaur26261 ziaur26261 4096 Sep 30 17:43 ..
-rw-r--r--  1 ziaur26261 ziaur26261 1327 Sep 30 17:43
webapps-2026-09-30_124302.tar.gz
-rw-r--r--  1 ziaur26261 ziaur26261  436 Sep 30 17:43
webapps-2026-09-30_124302.tar.gz.manifest
-rw-r--r--  1 ziaur26261 ziaur26261   99 Sep 30 17:43
webapps-2026-09-30_124302.tar.gz.sha256
-rw-r--r--  1 ziaur26261 ziaur26261 1637 Sep 30 18:27
webapps-2026-09-30_132750.tar.gz
-rw-r--r--  1 ziaur26261 ziaur26261  436 Sep 30 18:27
webapps-2026-09-30_132750.tar.gz.manifest
-rw-r--r--  1 ziaur26261 ziaur26261   99 Sep 30 18:27
webapps-2026-09-30_132750.tar.gz.sha256
-rw-r--r--  1 ziaur26261 ziaur26261 1637 Sep 30 18:29
webapps-2026-09-30_132943.tar.gz
-rw-r--r--  1 ziaur26261 ziaur26261  436 Sep 30 18:29
webapps-2026-09-30_132943.tar.gz.manifest
-rw-r--r--  1 ziaur26261 ziaur26261   99 Sep 30 18:29
webapps-2026-09-30_132943.tar.gz.sha256
```

06-vm-shape

```
### VM inventory, resources, and the snapshot used for rollback.
```

```
### captured 2026-09-30T13:40:09Z
```

```
$ virsh list --all
```

Id	Name	State
2	web01	running
3	app01	running

```
$ virsh dominfo app01
```

```
Id: 3
Name: app01
UUID: c5a5624f-3183-4664-8fd4-7b6df2950699
OS Type: hvm
State: running
CPU(s): 1
CPU time: 250.3s
Max memory: 4194304 KiB
Used memory: 2097152 KiB
Persistent: yes
Autostart: enable
Managed save: no
Security model: none
Security DOI: 0
```

```
$ virsh snapshot-list app01
```

Name	Creation Time	State
clean-deploy	2026-09-30 18:38:28 +0500	running

```
$ virsh net-dhcp-leases default
```

Expiry Time	MAC address	Protocol	IP address	Hostname
-------------	-------------	----------	------------	----------

```
Client ID or DUID
```

2026-09-30 19:33:02	52:54:00:aa:00:11	ipv4	192.168.122.11/24	app01
ff:56:50:4d:98:00:02:00:00:ab:11:21:76:16:bf:90:fc:10:55				
2026-09-30 19:37:11	52:54:00:aa:00:12	ipv4	192.168.122.12/24	web01
ff:56:50:4d:98:00:02:00:00:ab:11:1c:67:f0:40:53:75:82:af				

```
$ ssh ops@192.168.122.11 'hostname; nproc; free -m'
```

```
app01
```

	total	used	free	shared	buff/cache	available
Mem:	1867	499	592	5	1020	1367
Swap:	0	0	0			

07-live-resize

```
### Live CPU and memory resize, and the failure mode where a hot-added vCPU
stays offline until the guest is told to online it.
### captured 2026-09-30T13:40:10Z

$ virsh dominfo app01 | grep -E '^CPU\(s\)|Used memory'      # before
CPU(s):              1
Used memory:         2097152 KiB
$ ssh ops@192.168.122.11 'nproc'
1

$ virsh setvcpus app01 2 --live; virsh setmem app01 3G --live

    setvcpus ok

    setmem ok

$ ssh ops@192.168.122.11 'nproc'      # still 1: ACPI hot-added CPU1 and left it
offline
1
0-1
0

$ ssh ops@192.168.122.11 'echo 1 | sudo tee /sys/devices/system/cpu/cpu1/online;
nproc'
2

$ virsh setmem app01 3G --live  -> visible inside the guest
Mem:           2891           535           1580           5           1020           2355

$ virsh setvcpus app01 5 --live      # past the maximum fixed at creation
error: invalid argument: requested vcpus is greater than max allowable vcpus for the
live domain: 5 > 4

$ virsh setmem app01 8G --live
error: invalid argument: cannot set memory higher than max memory

# restoring the documented shape
```

08-monitoring

```
### Prometheus scrape targets, loaded alert rules, and Grafana health.
### captured 2026-09-30T13:40:15Z

$ curl -s localhost:9090/api/v1/targets
192.168.122.11:9100    job=node    health=up
192.168.122.12:9100    job=node    health=up

$ curl -s localhost:9090/api/v1/rules
InstanceDown    state=inactive
HighCPU          state=inactive
LowDisk          state=inactive
LowMemory        state=inactive

$ curl -s localhost:3000/api/health
{
  "database": "ok",
  "version": "13.2.3",
  "commit": "90ffed056f0884267356c12a0eeb72a022af53f1"
}
$ curl -s -u admin:admin 'localhost:3000/api/search?type=dash-db'
Node Exporter Full (uid=rYdddLPWk)
```